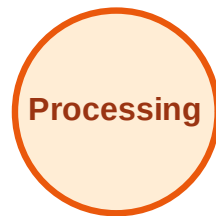


BFS Traversal

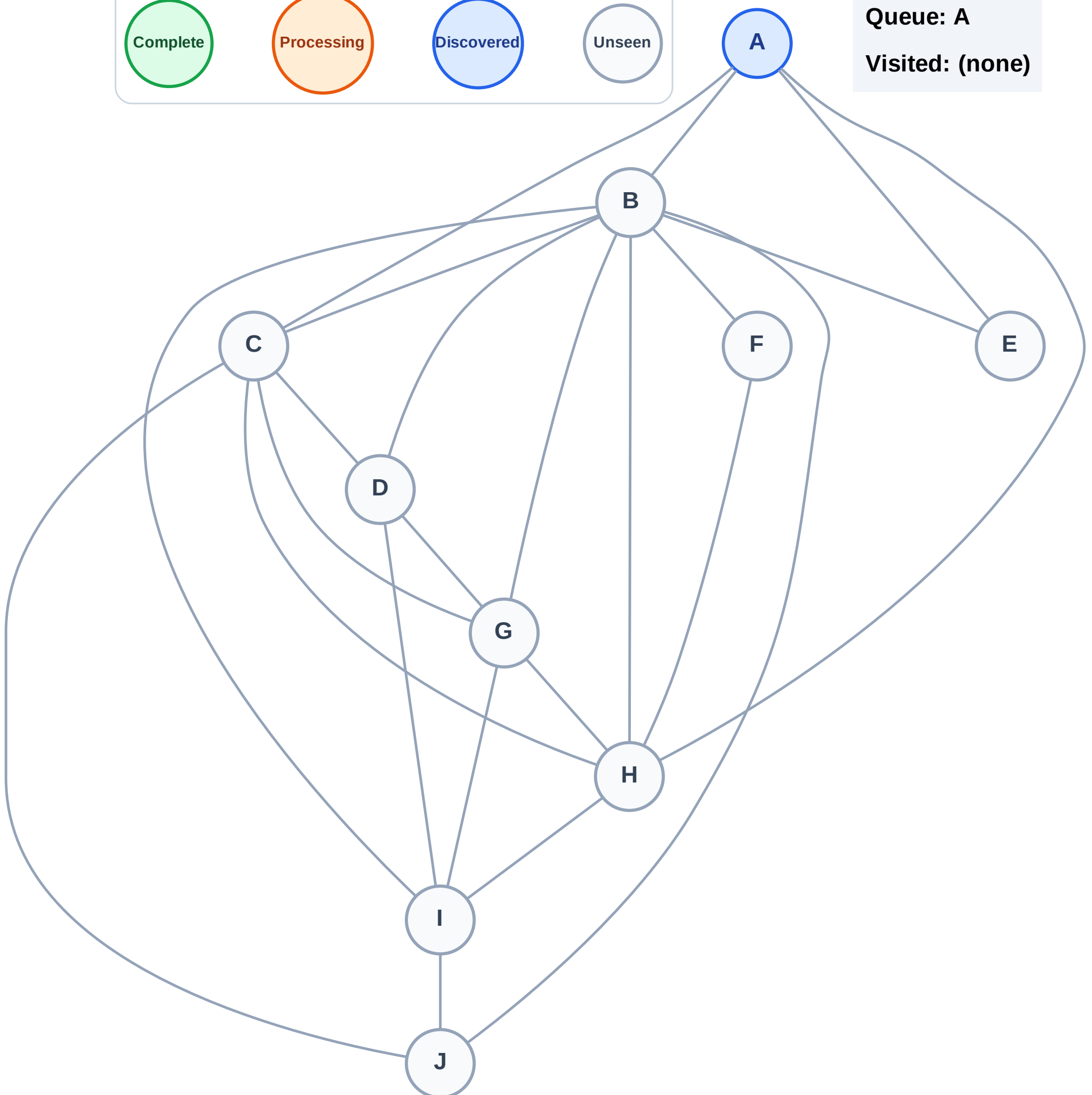
Step 1: Start at A

Legend



Queue: A

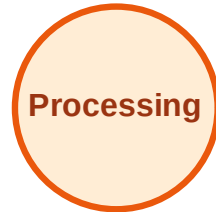
Visited: (none)



BFS Traversal

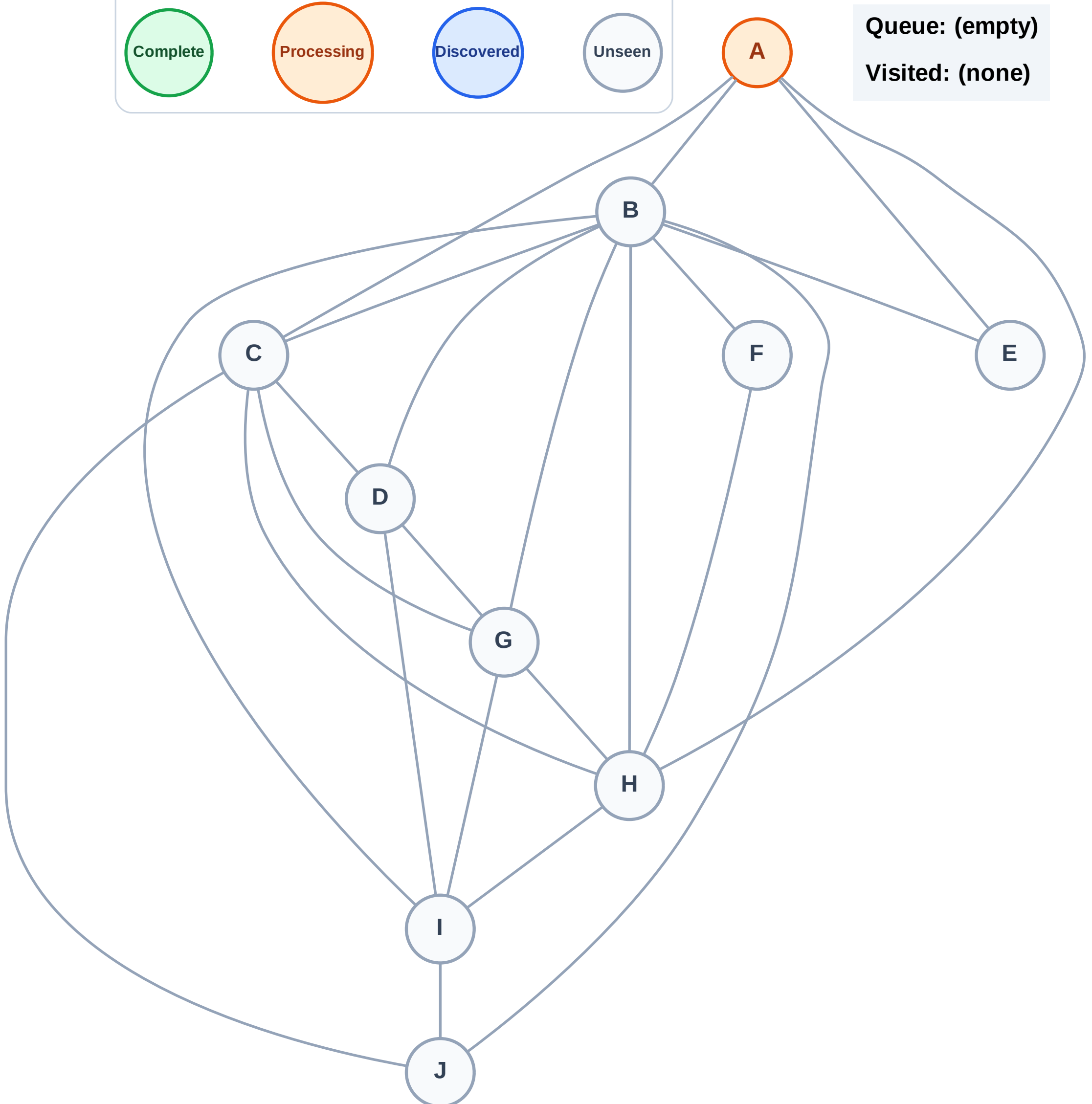
Step 2: Process node A

Legend



Queue: (empty)

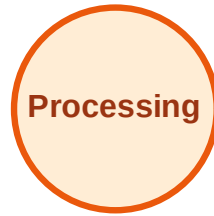
Visited: (none)



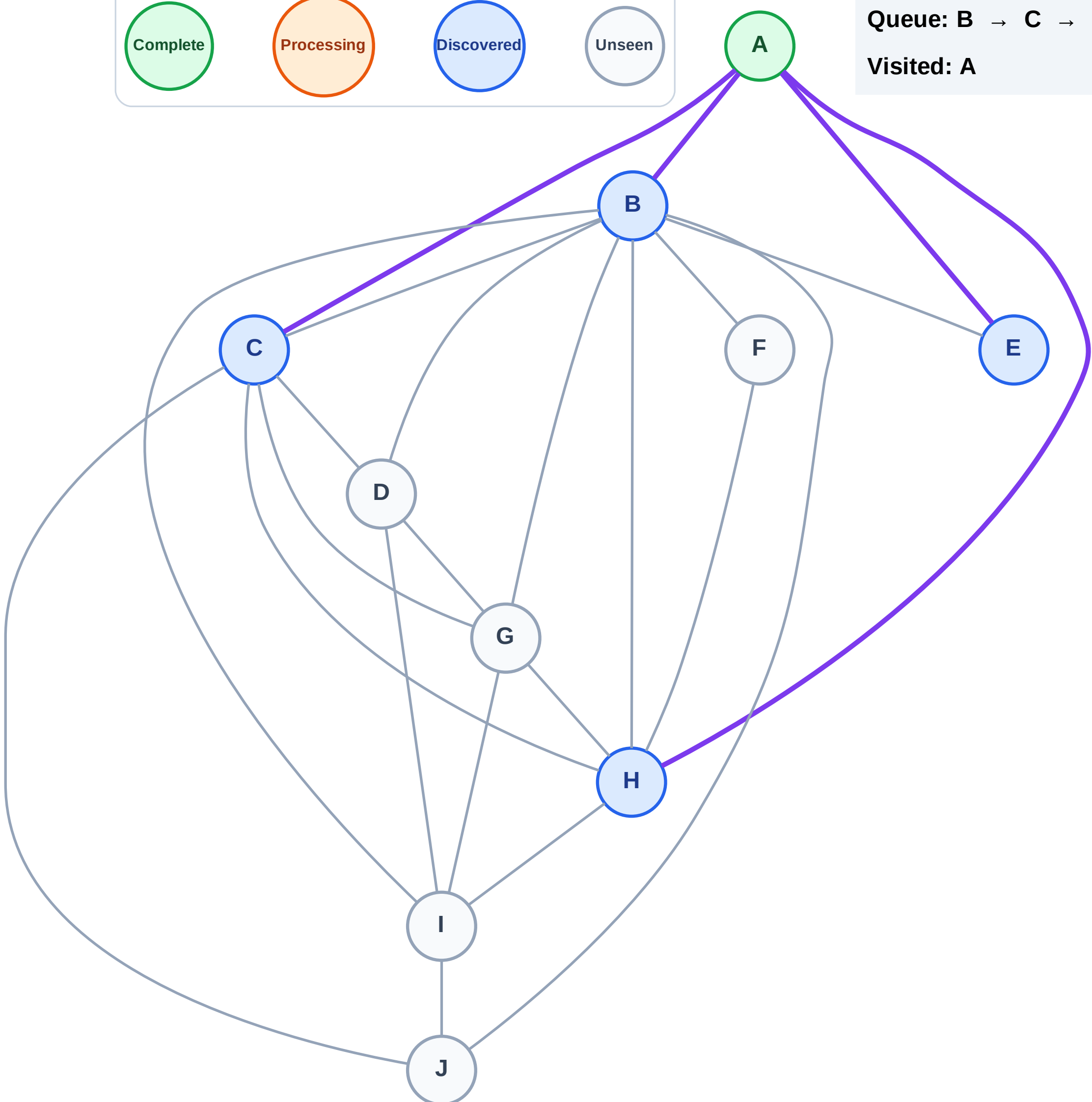
BFS Traversal

Step 3: Discover B, C, E, H from A

Legend



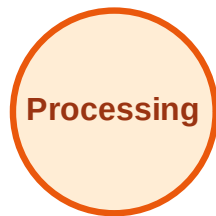
Queue: B → C → E → H
Visited: A



BFS Traversal

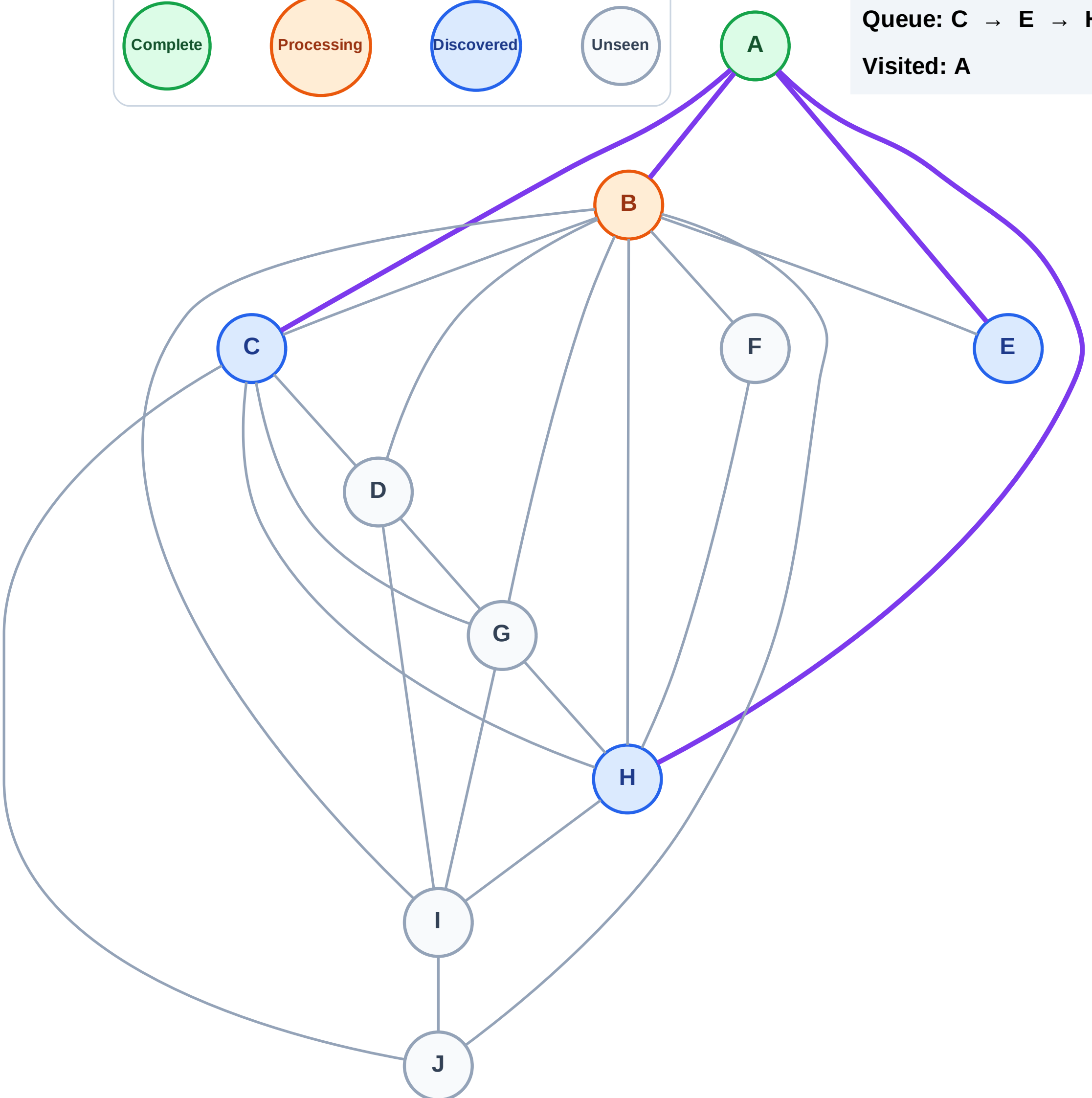
Step 4: Process node B

Legend



Queue: C → E → H

Visited: A



BFS Traversal

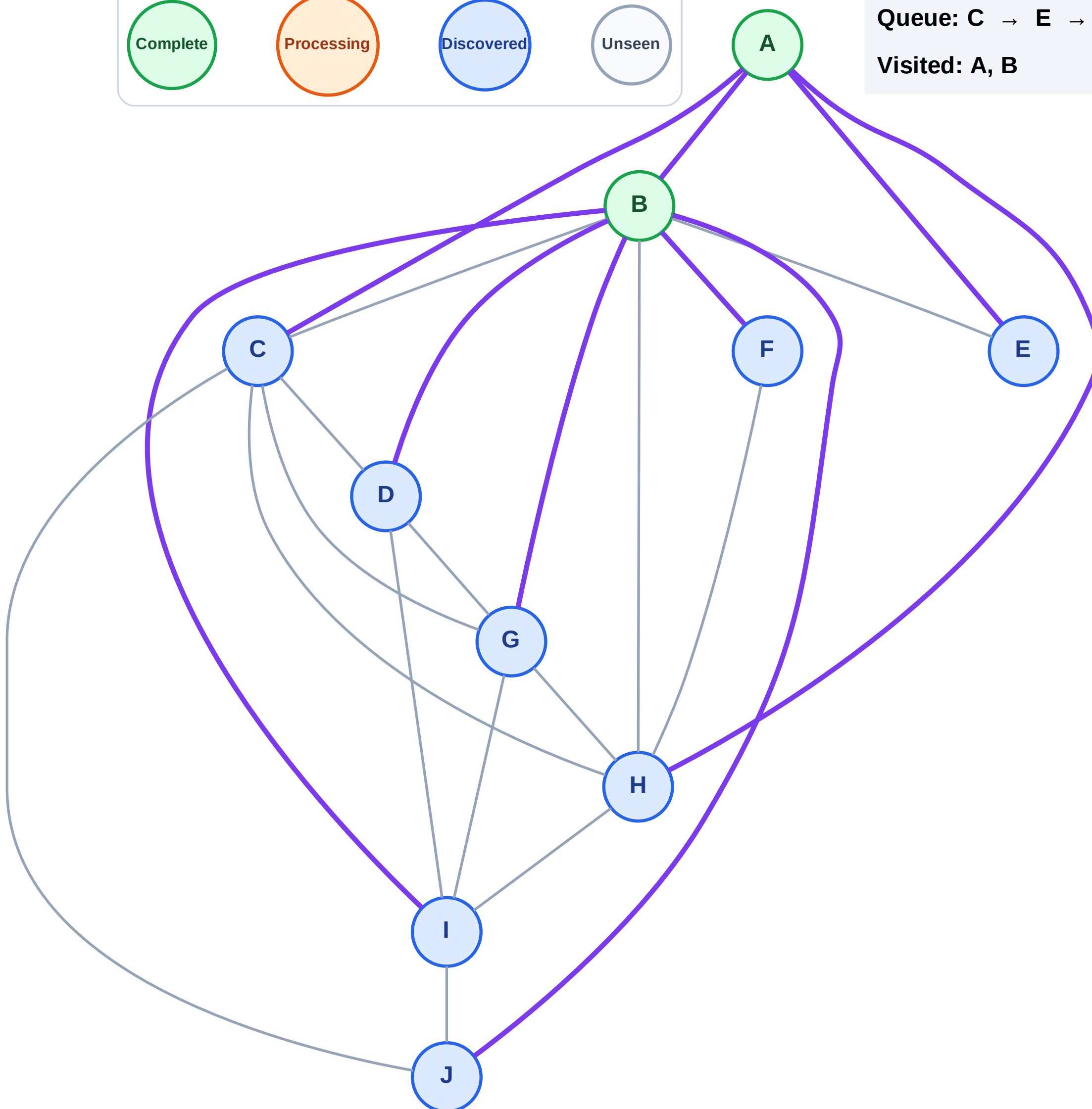
Step 5: Discover D, F, G, I, J from B

Legend



Queue: C → E → H → D → F → G → I → J

Visited: A, B



BFS Traversal

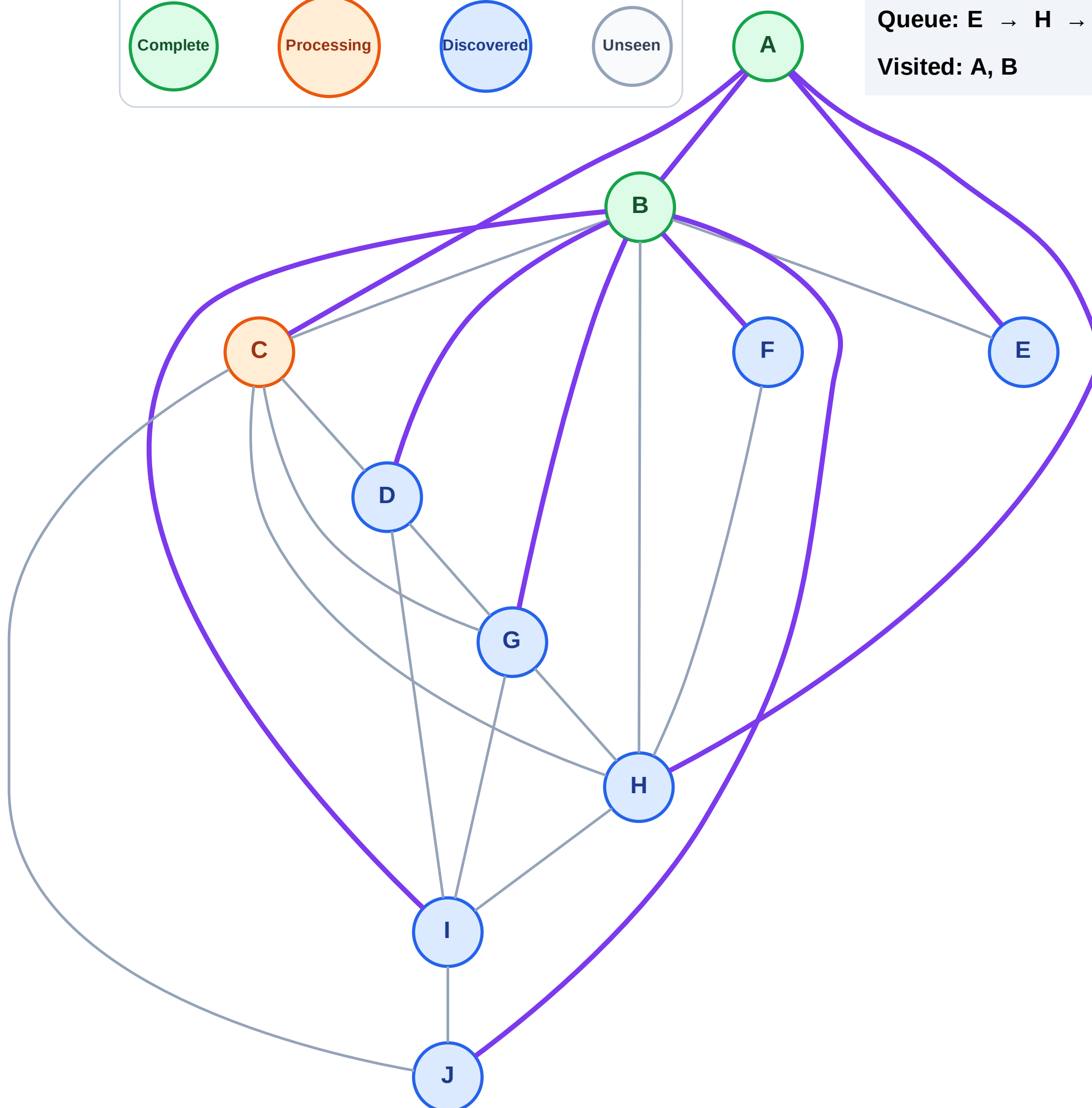
Step 6: Process node C

Legend



Queue: E → H → D → F → G → I → J

Visited: A, B



BFS Traversal

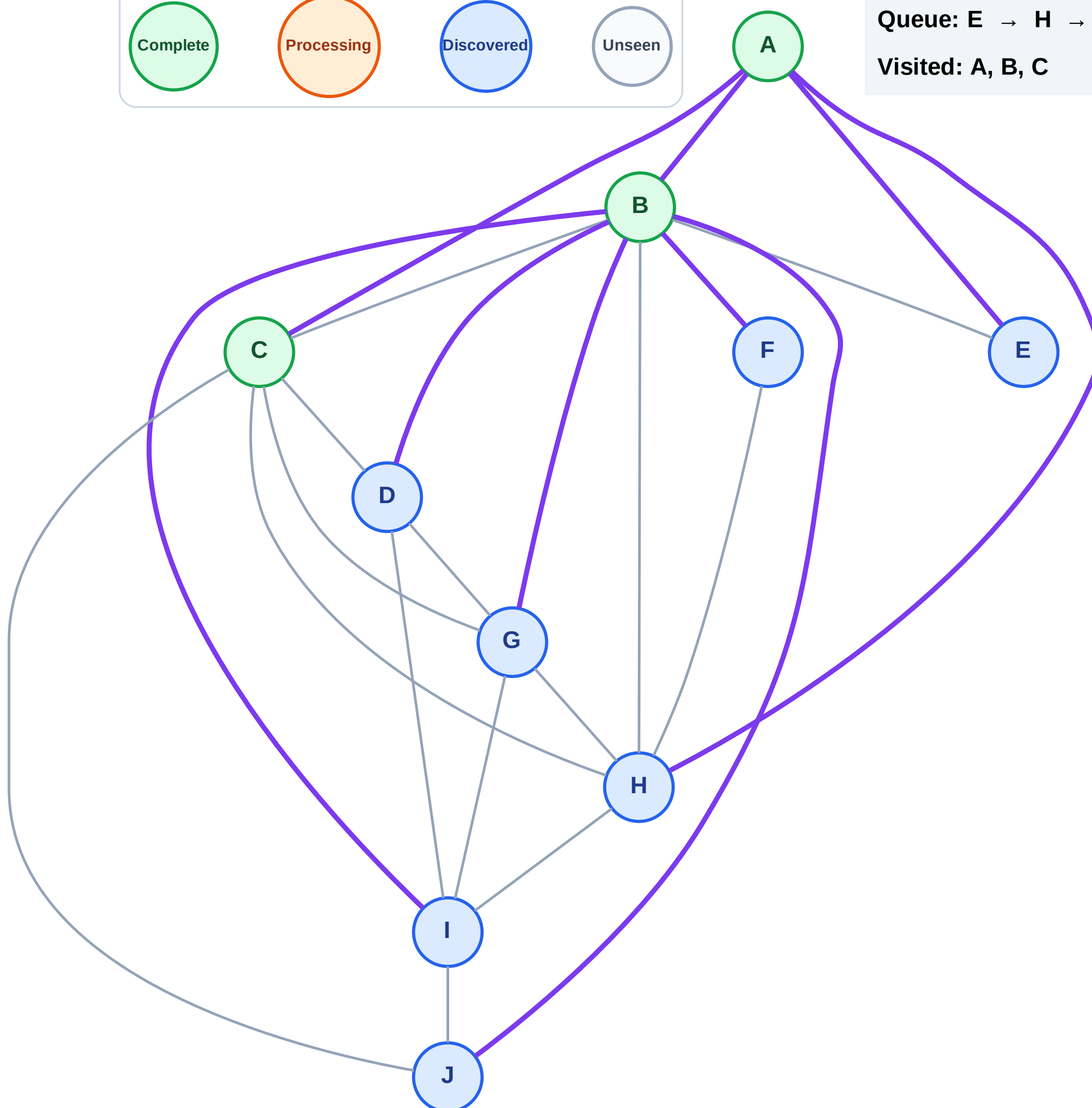
Step 7: No new neighbors from C

Legend



Queue: E → H → D → F → G → I → J

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

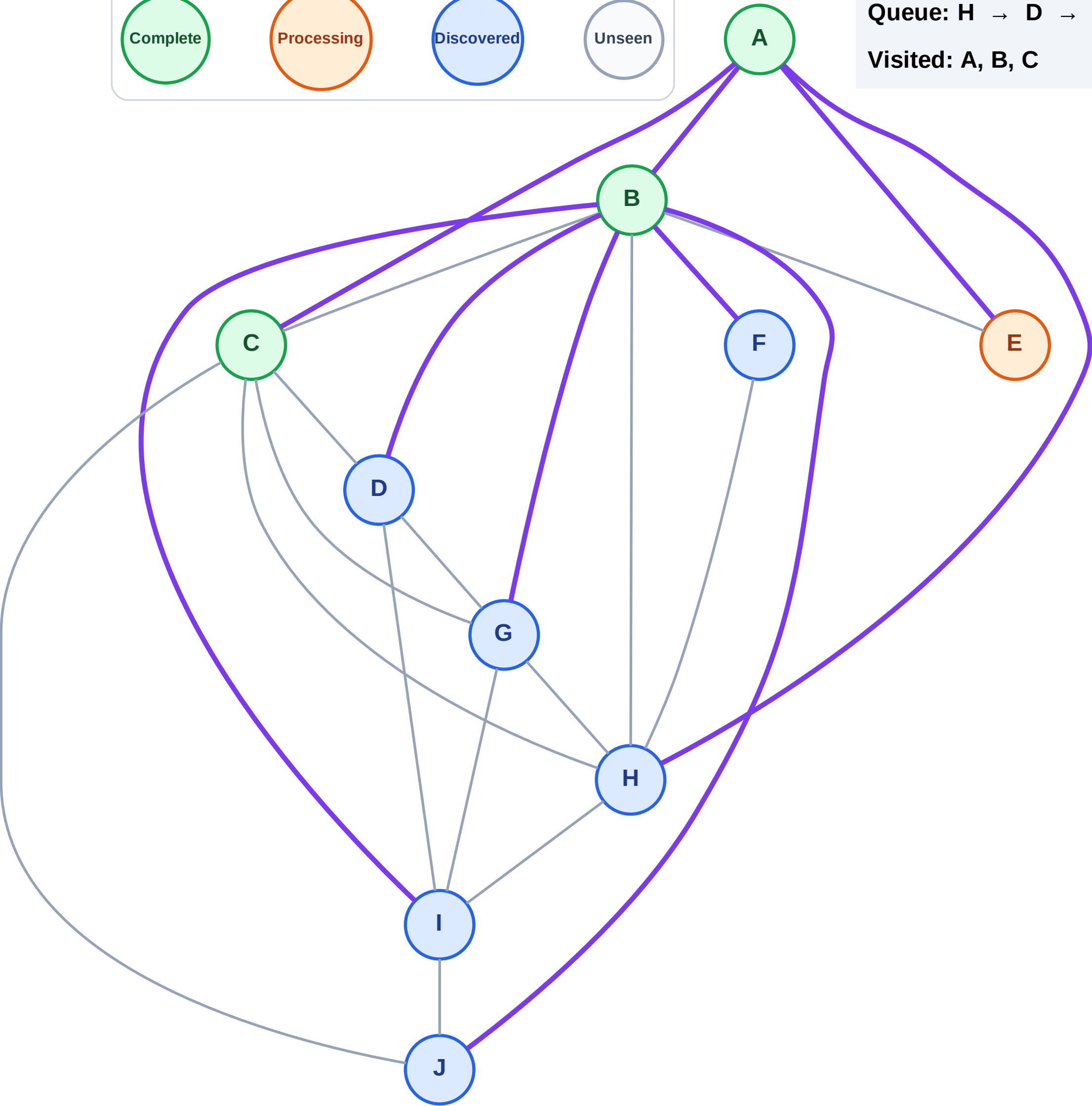
Processing

Discovered

Unseen

Queue: H → D → F → G → I → J

Visited: A, B, C



BFS Traversal

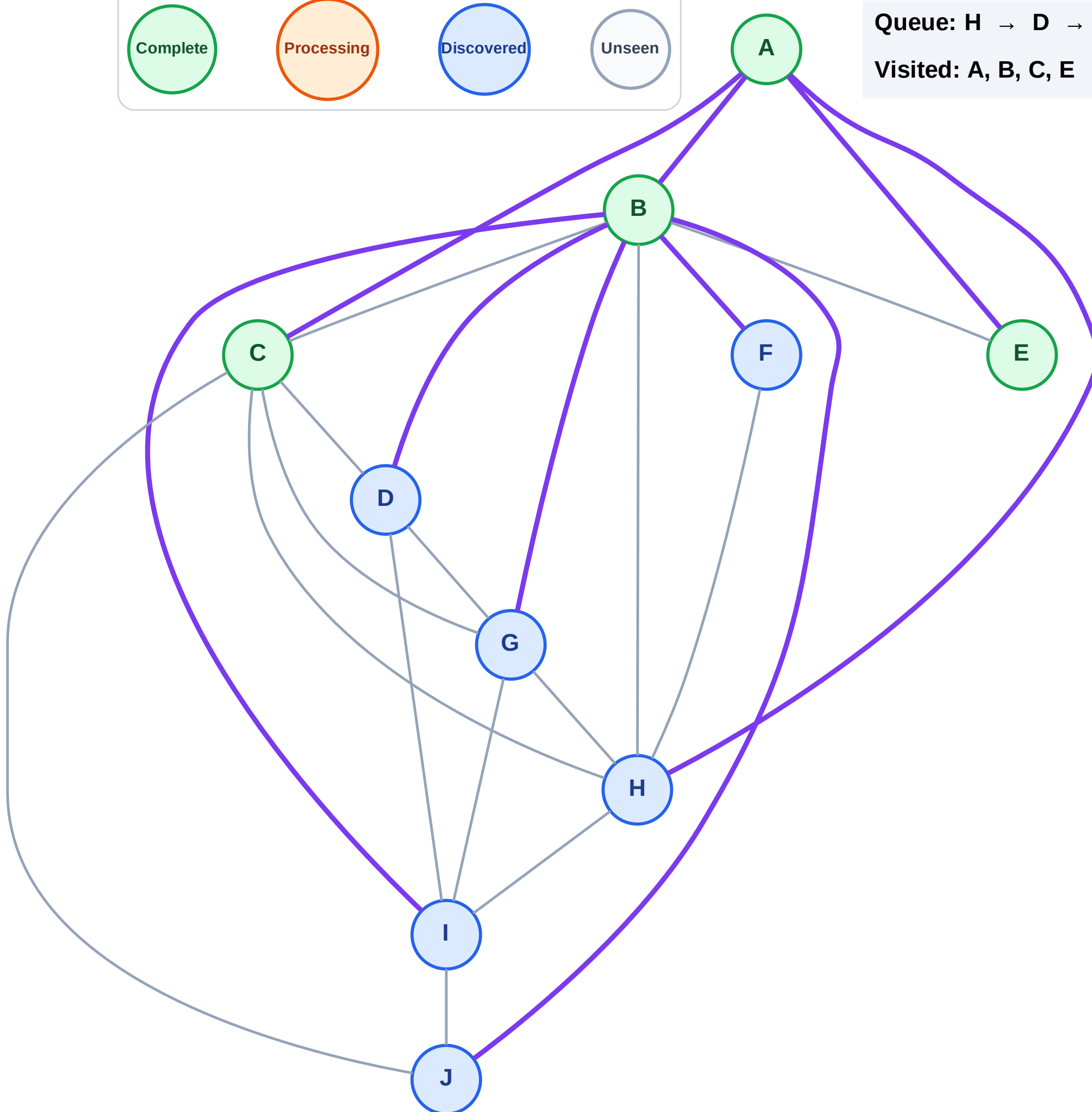
Step 9: No new neighbors from E

Legend



Queue: H → D → F → G → I → J

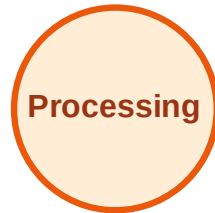
Visited: A, B, C, E



BFS Traversal

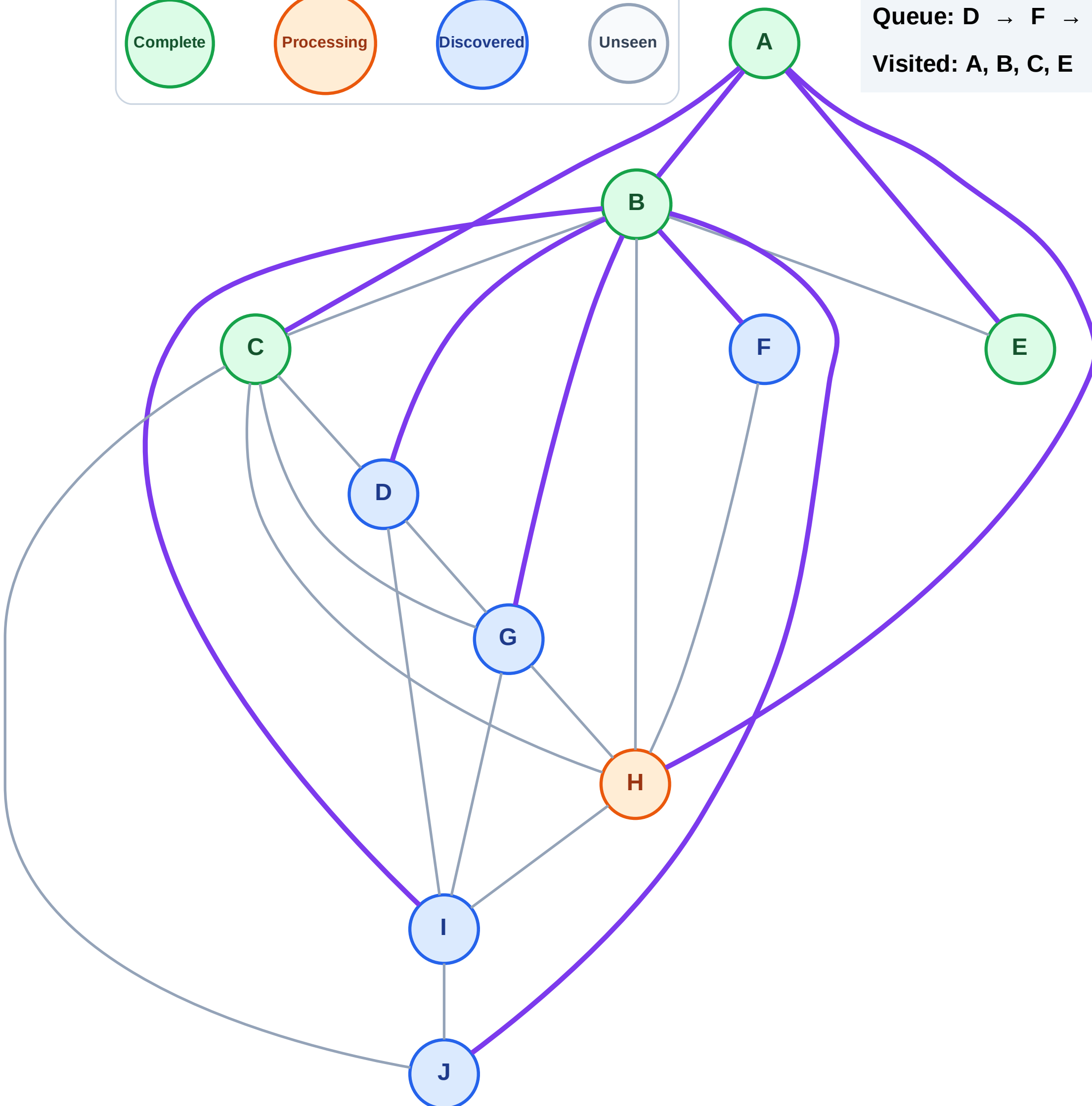
Step 10: Process node H

Legend



Queue: D → F → G → I → J

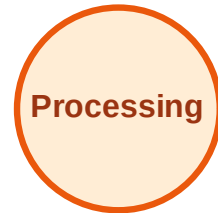
Visited: A, B, C, E



BFS Traversal

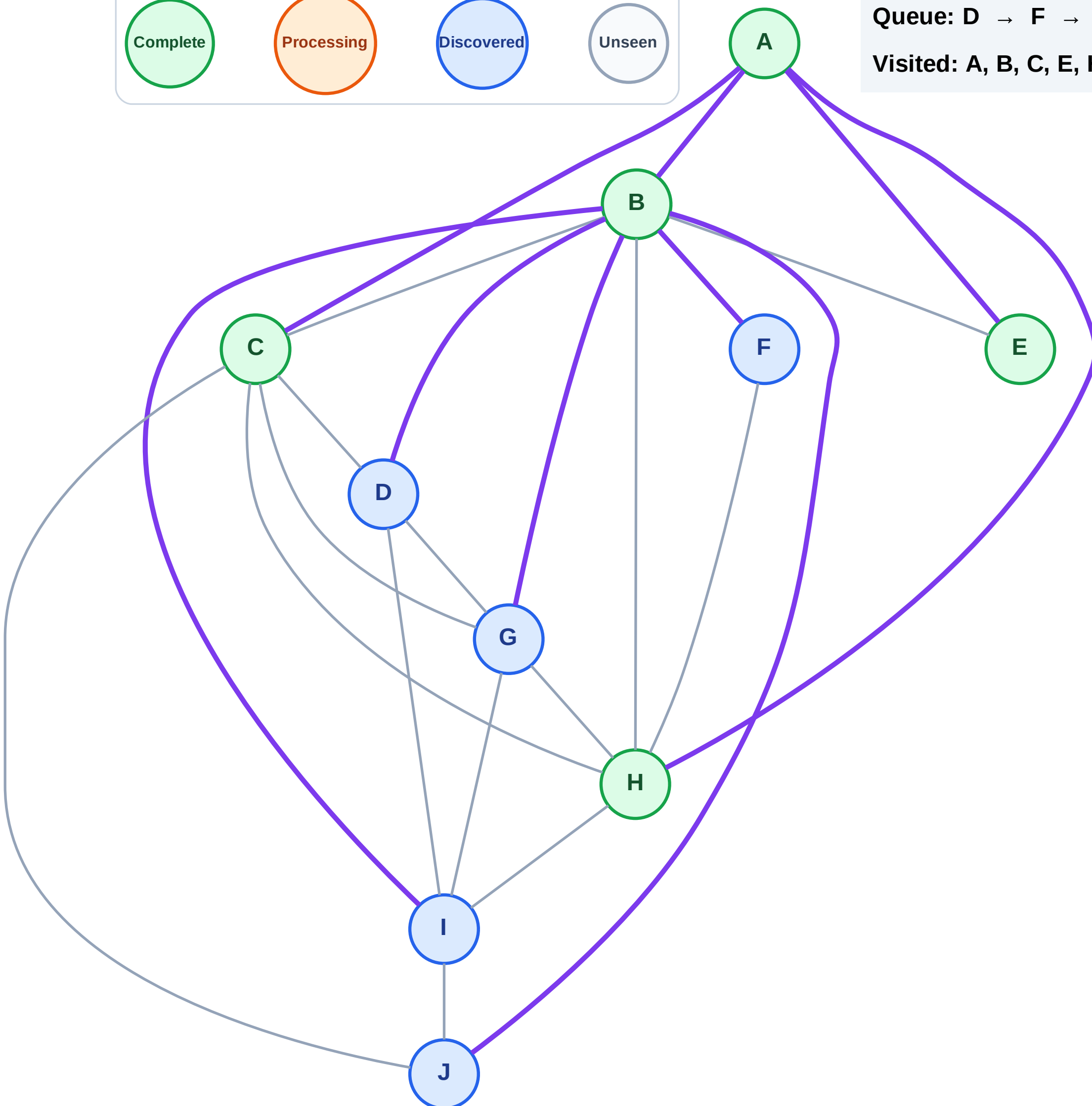
Step 11: No new neighbors from H

Legend



Queue: D → F → G → I → J

Visited: A, B, C, E, H



BFS Traversal

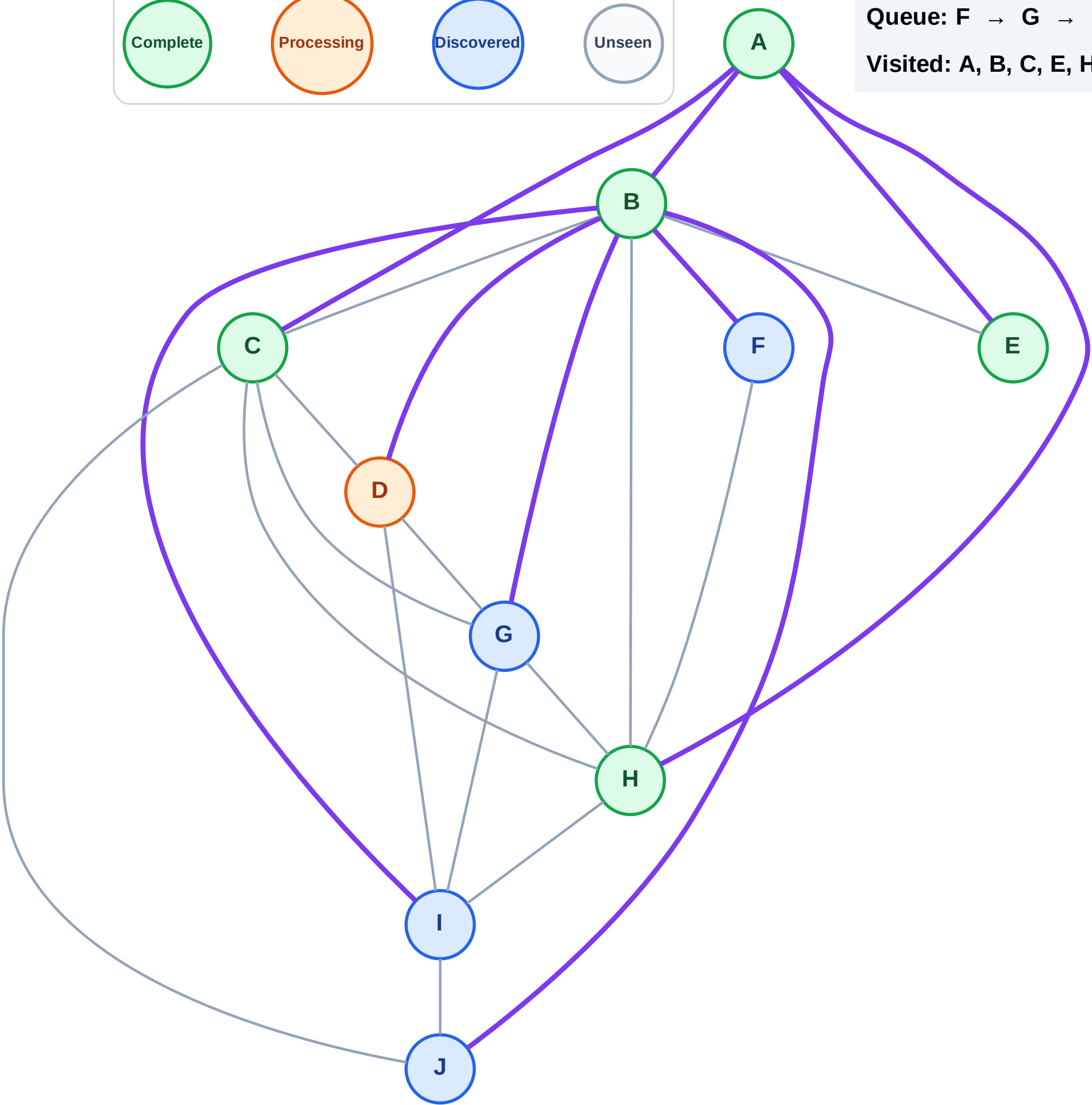
Step 12: Process node D

Legend



Queue: F → G → I → J

Visited: A, B, C, E, H



BFS Traversal

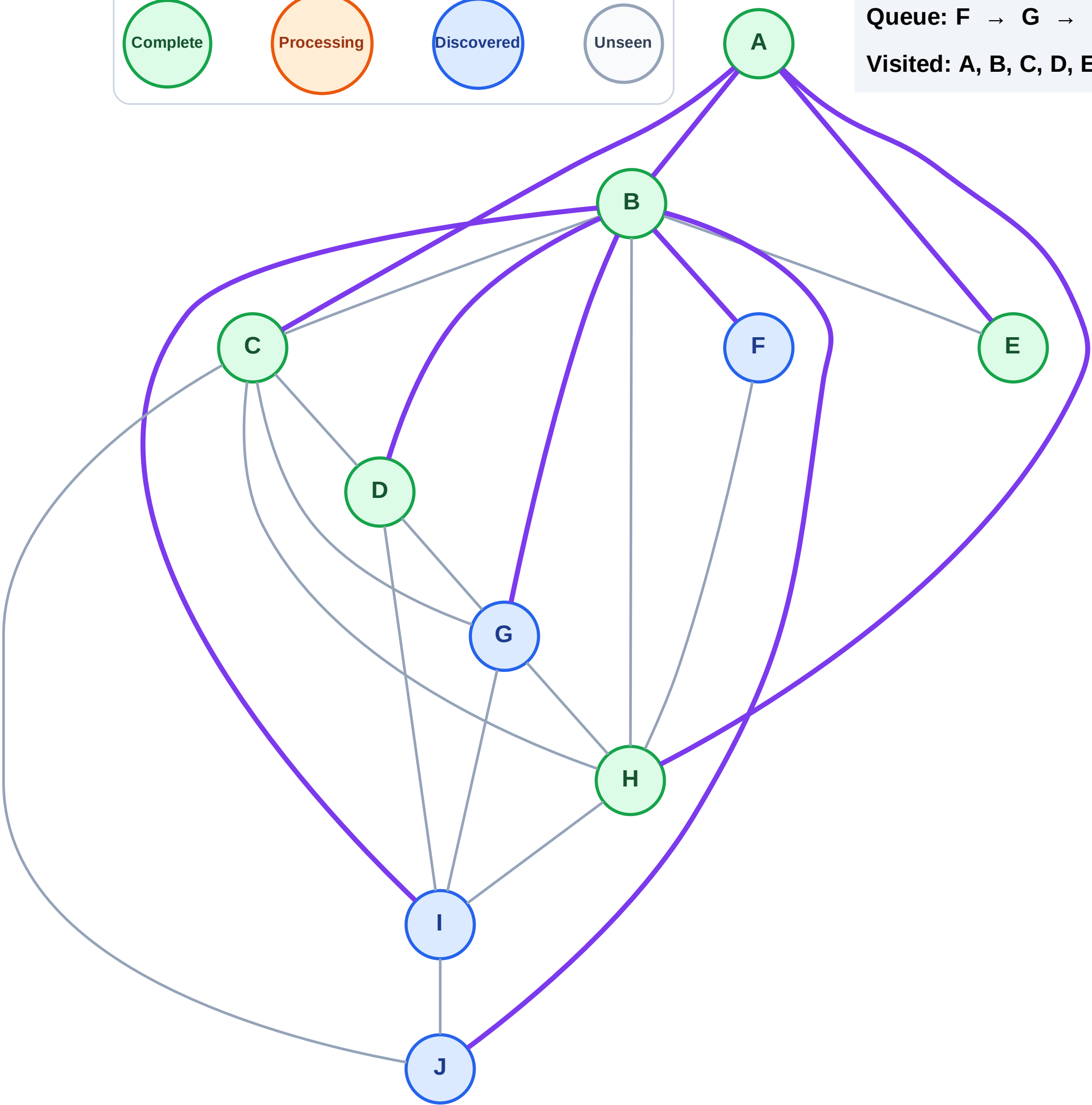
Step 13: No new neighbors from D

Legend



Queue: F → G → I → J

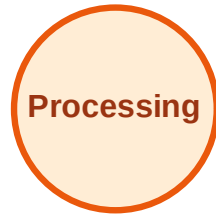
Visited: A, B, C, D, E, H



BFS Traversal

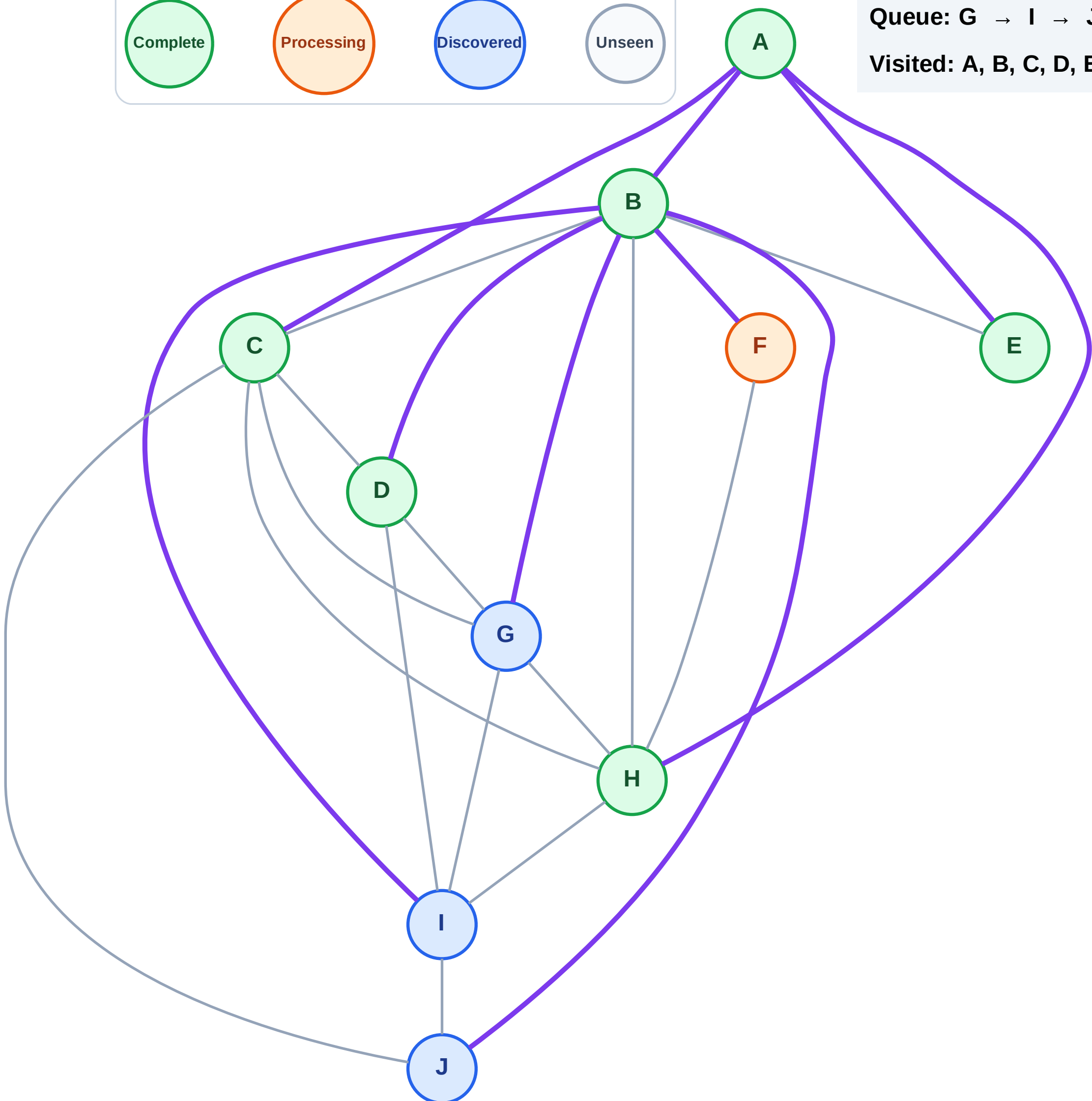
Step 14: Process node F

Legend



Queue: G → I → J

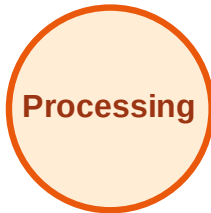
Visited: A, B, C, D, E, H



BFS Traversal

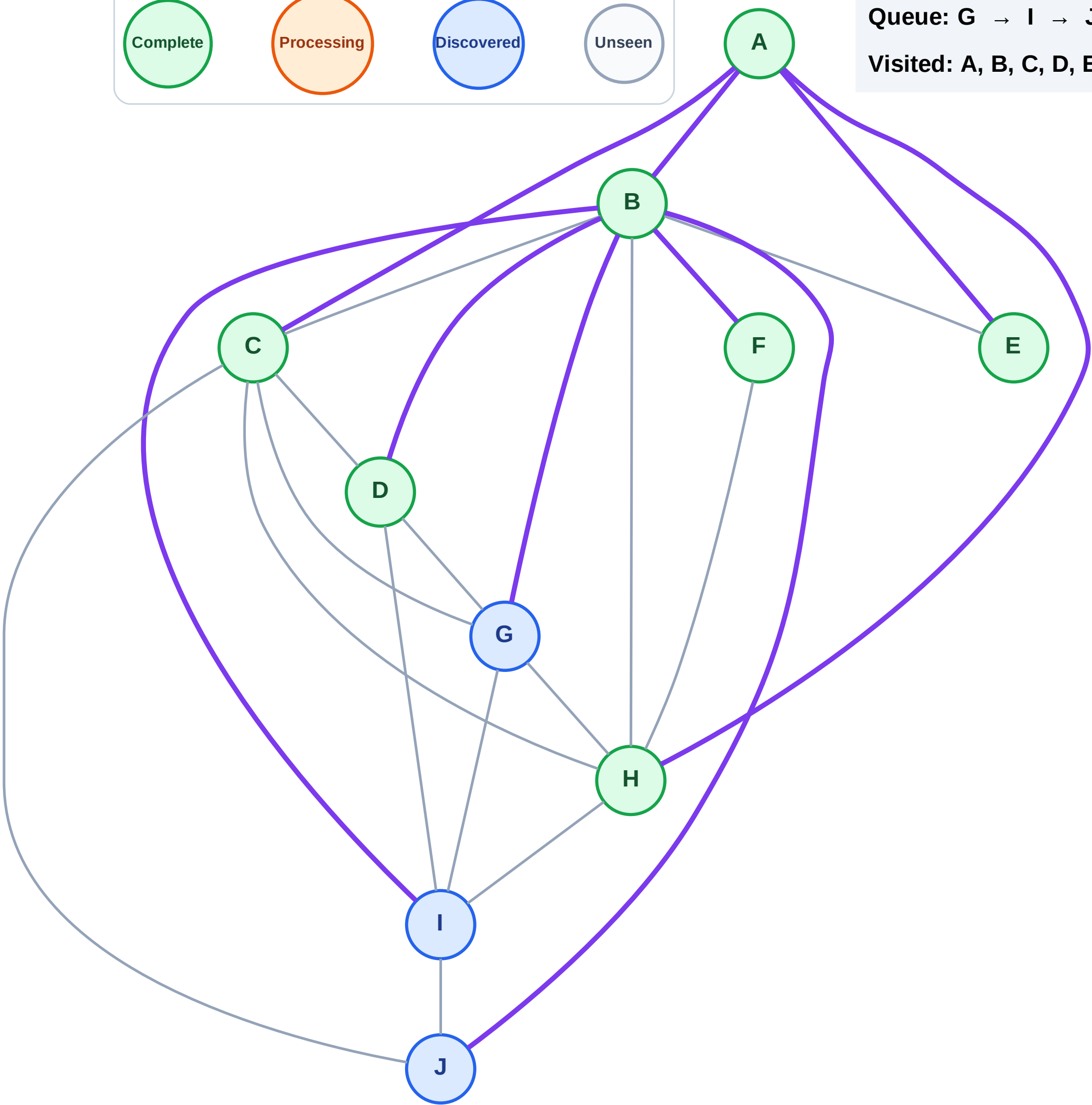
Step 15: No new neighbors from F

Legend



Queue: G → I → J

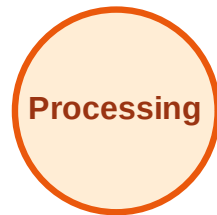
Visited: A, B, C, D, E, F, H



BFS Traversal

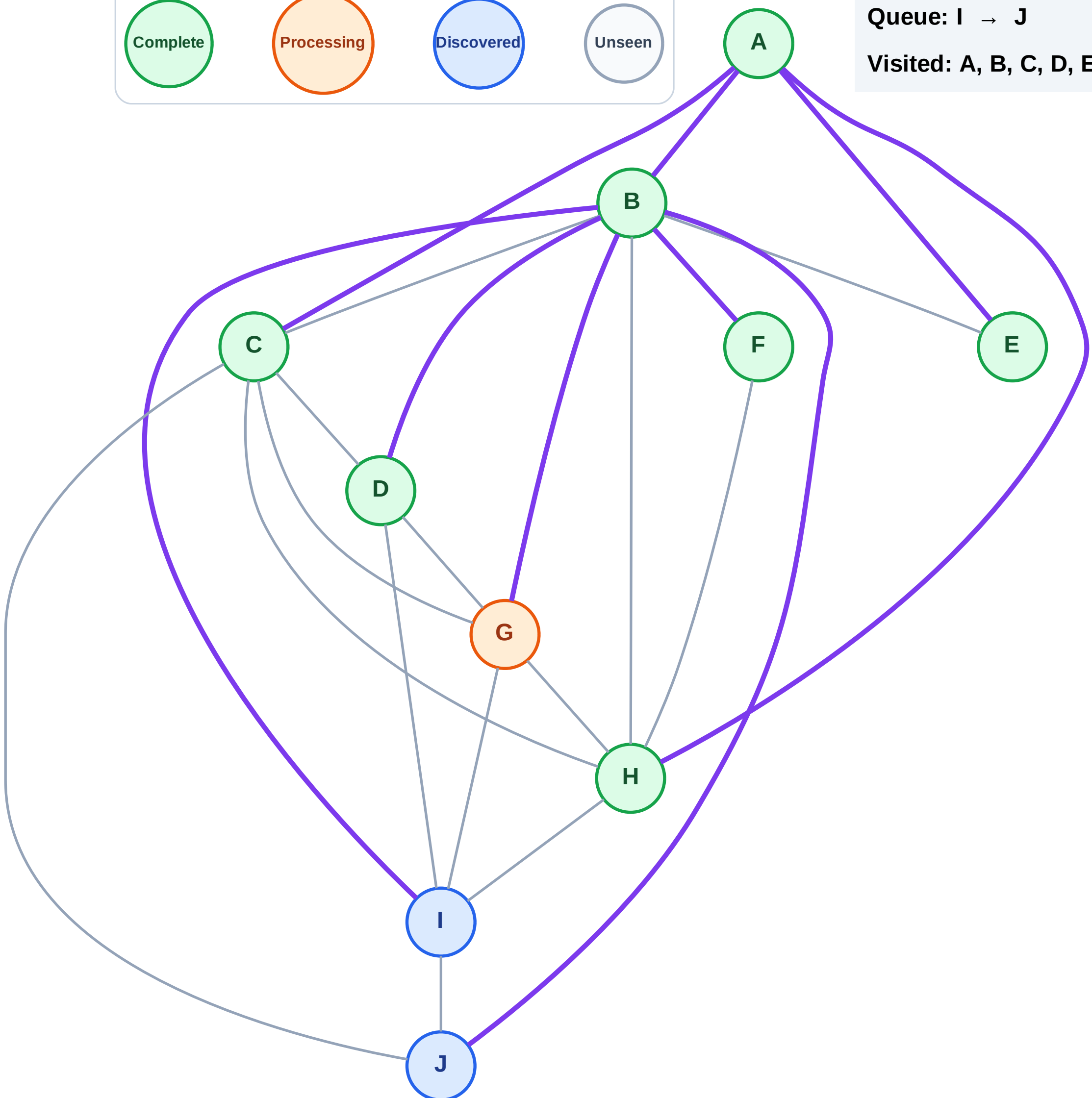
Step 16: Process node G

Legend



Queue: I → J

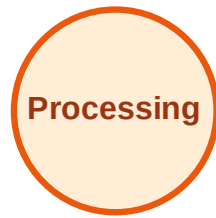
Visited: A, B, C, D, E, F, H



BFS Traversal

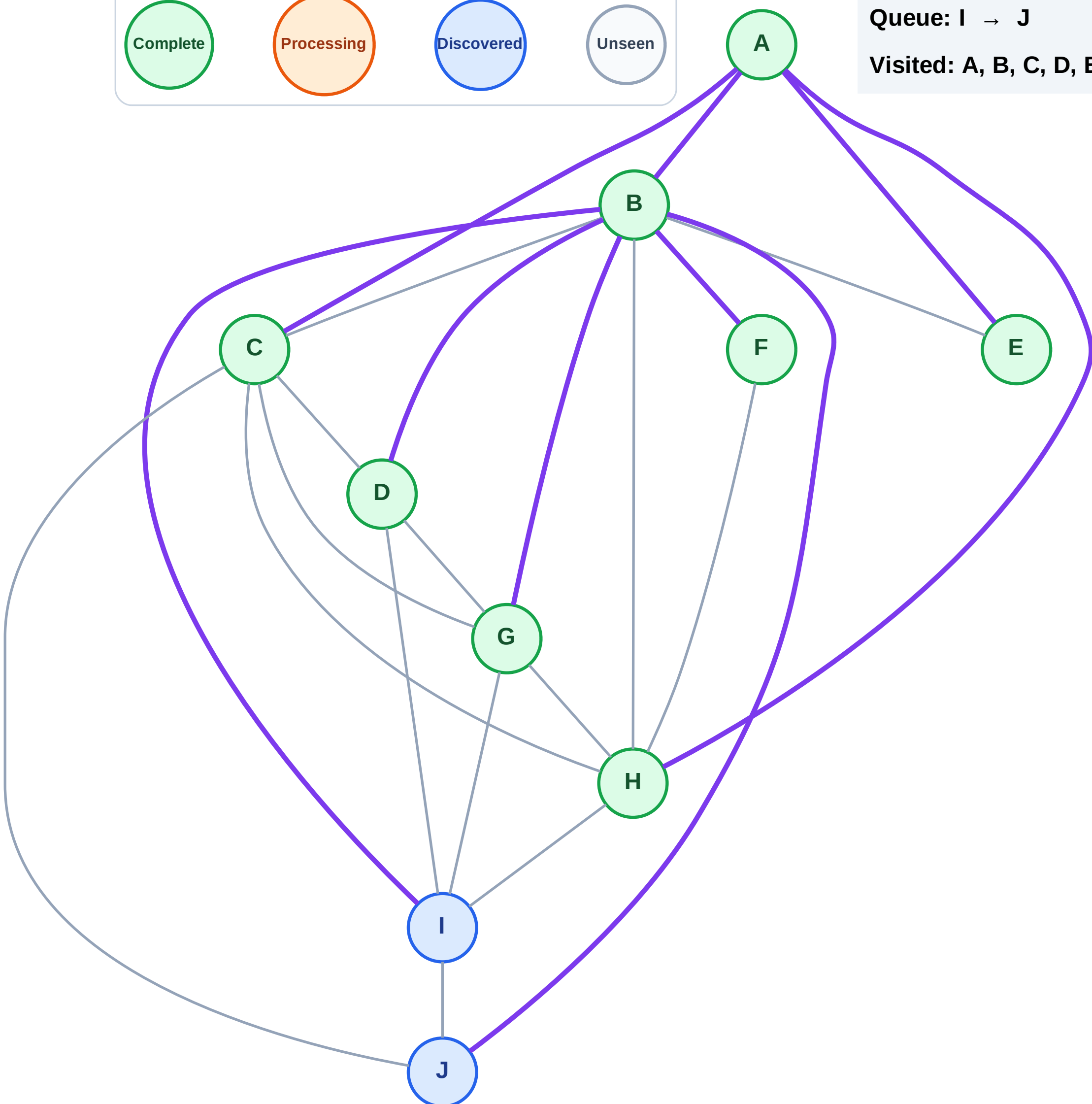
Step 17: No new neighbors from G

Legend



Queue: I → J

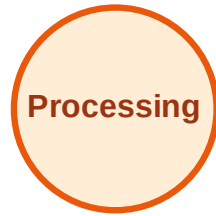
Visited: A, B, C, D, E, F, G, H



BFS Traversal

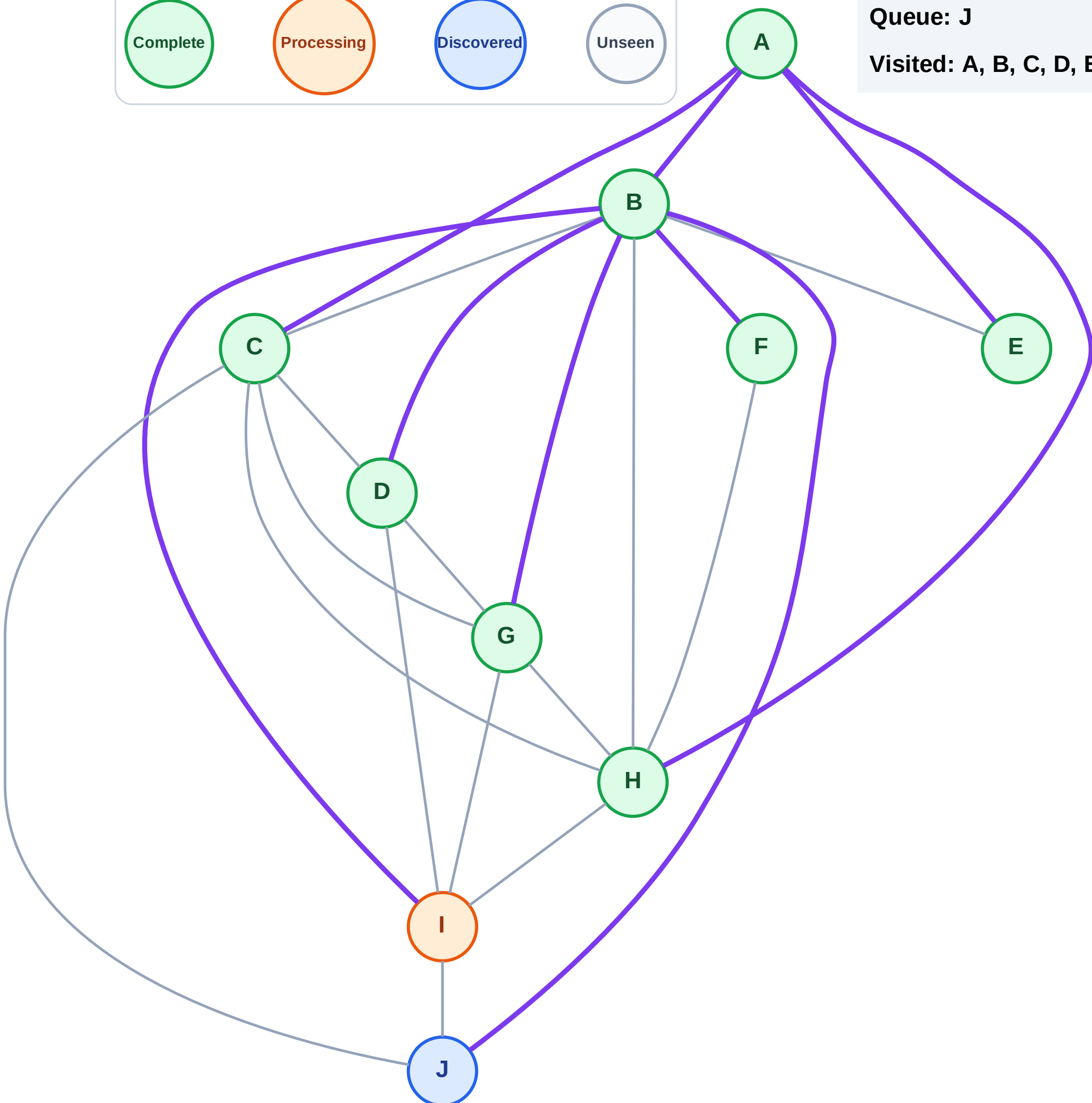
Step 18: Process node I

Legend



Queue: J

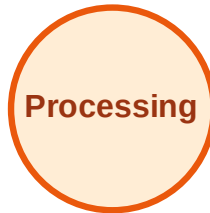
Visited: A, B, C, D, E, F, G, H



BFS Traversal

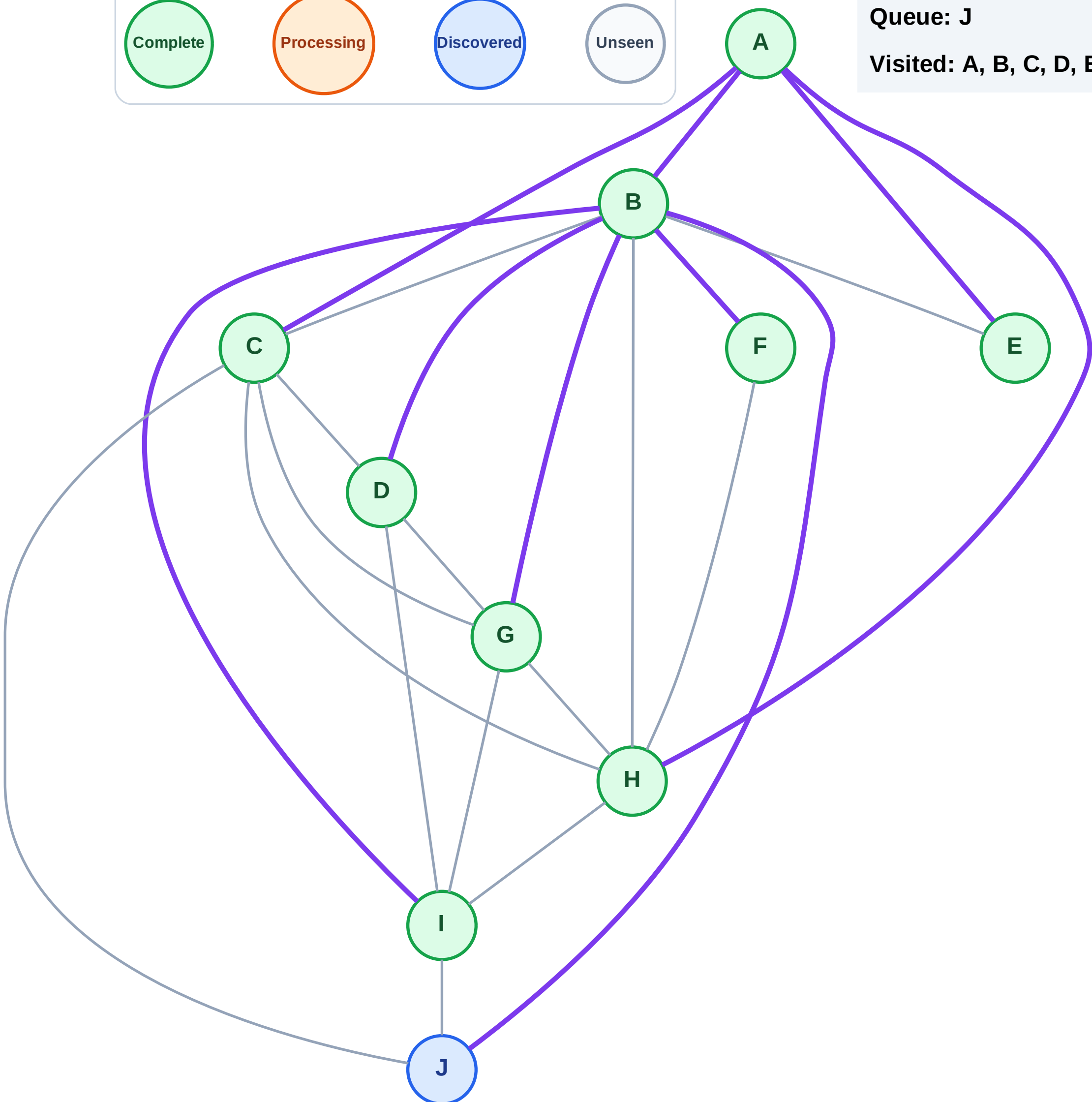
Step 19: No new neighbors from I

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

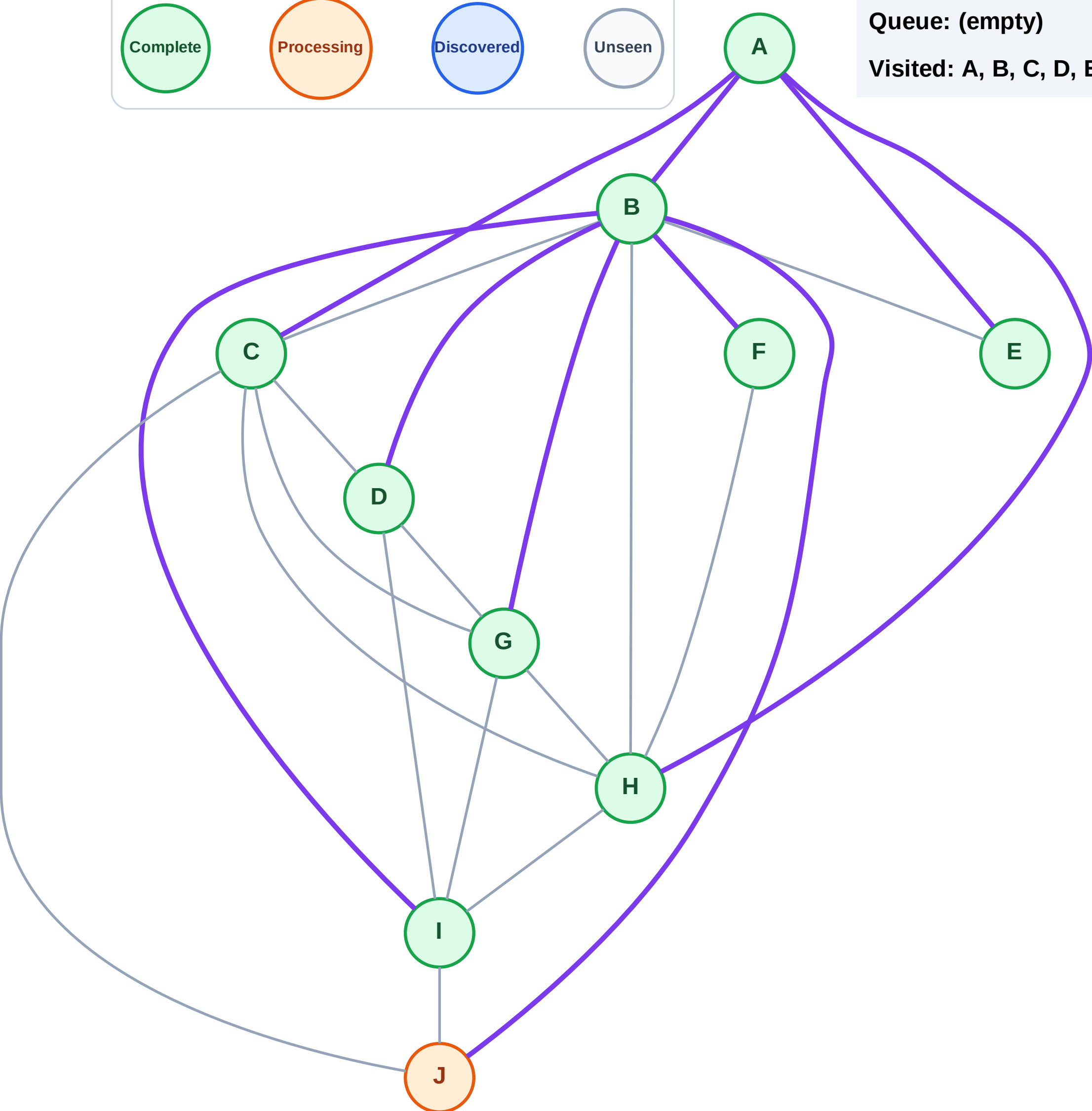
Step 20: Process node J

Legend



Queue: (empty)

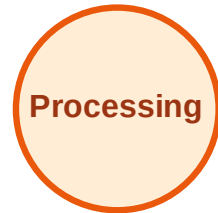
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

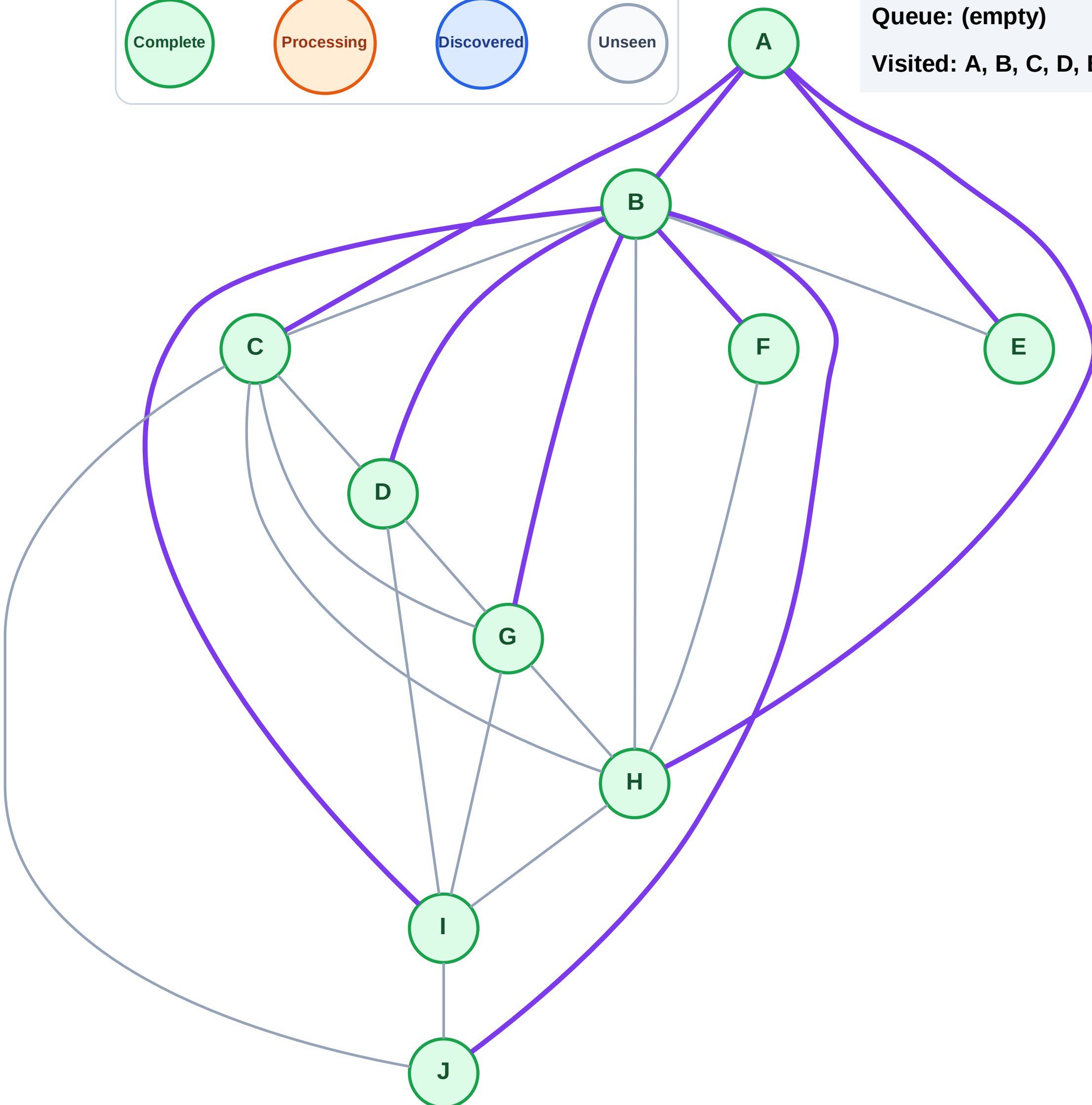
Step 21: No new neighbors from J

Legend



Queue: (empty)

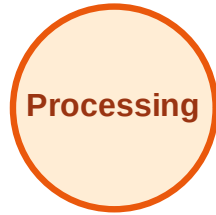
Visited: A, B, C, D, E, F, G, H, I, J



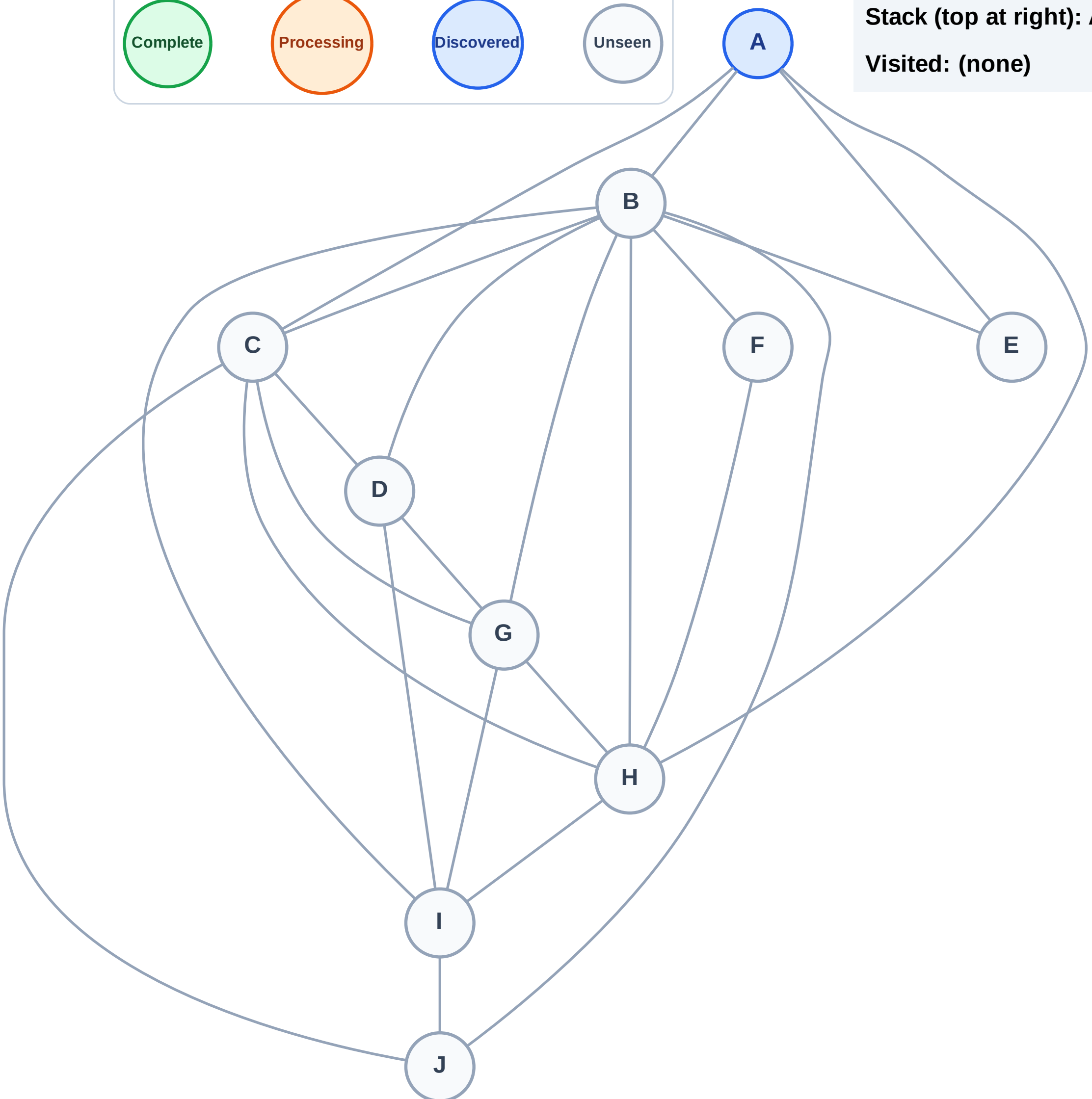
DFS Traversal

Step 1: Start at A

Legend



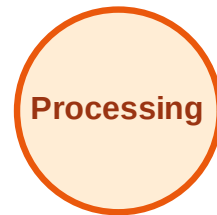
Stack (top at right): A
Visited: (none)



DFS Traversal

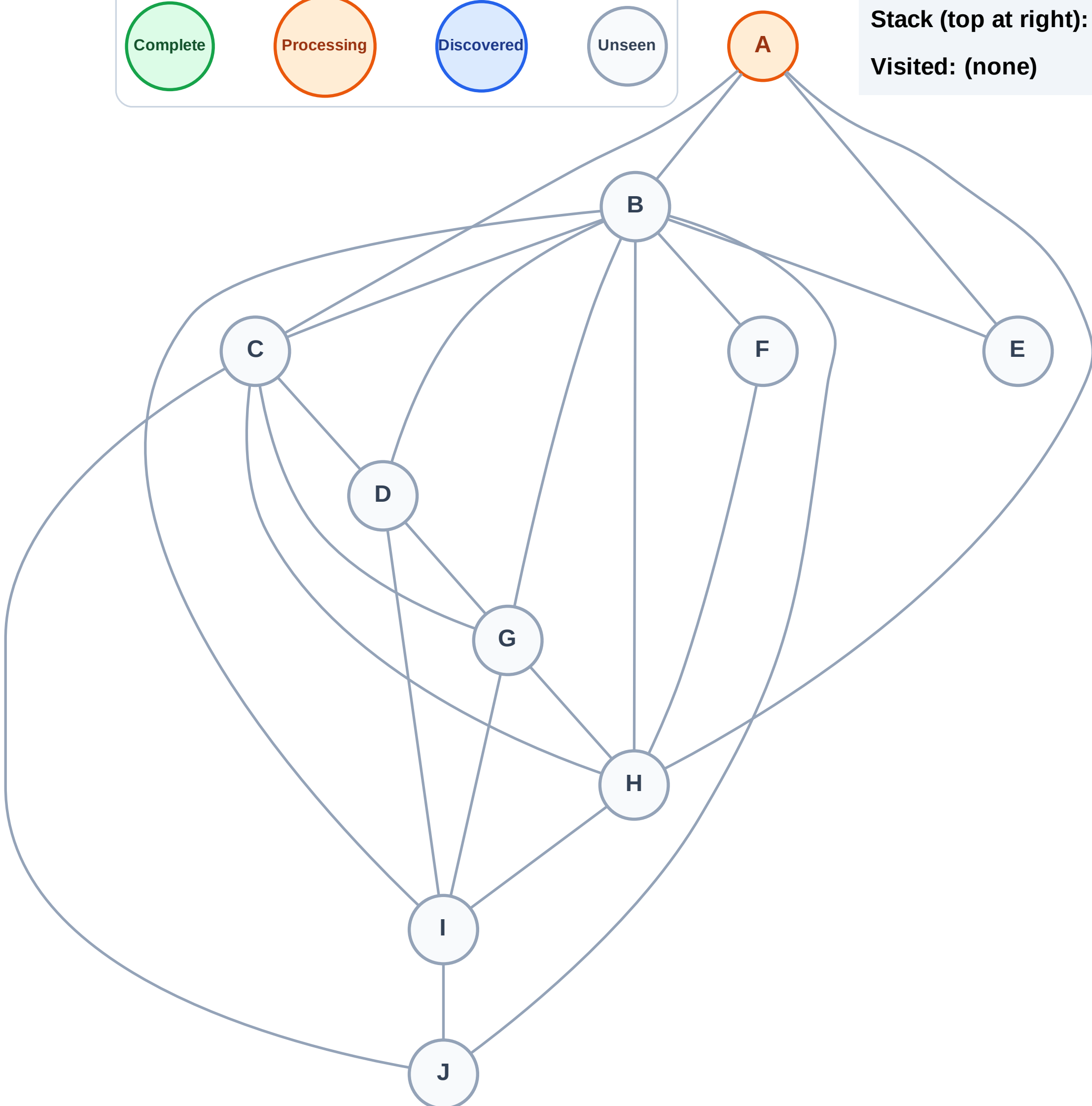
Step 2: Process node A

Legend



Stack (top at right): (empty)

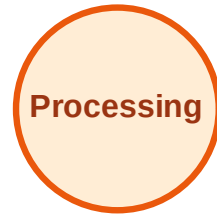
Visited: (none)



DFS Traversal

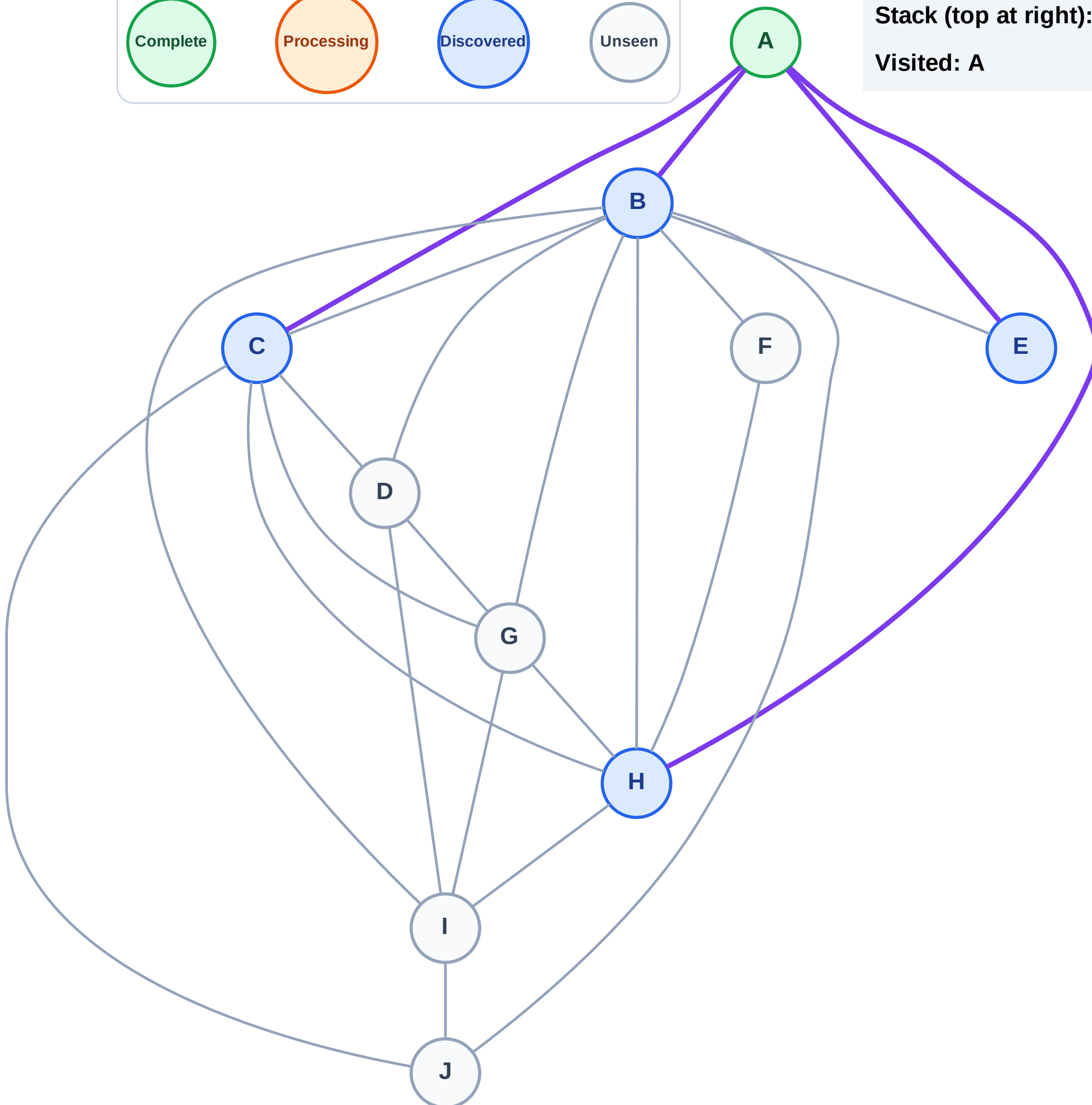
Step 3: Discover H, E, C, B from A

Legend



Stack (top at right): H | E | C | B

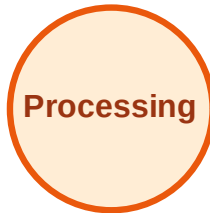
Visited: A



DFS Traversal

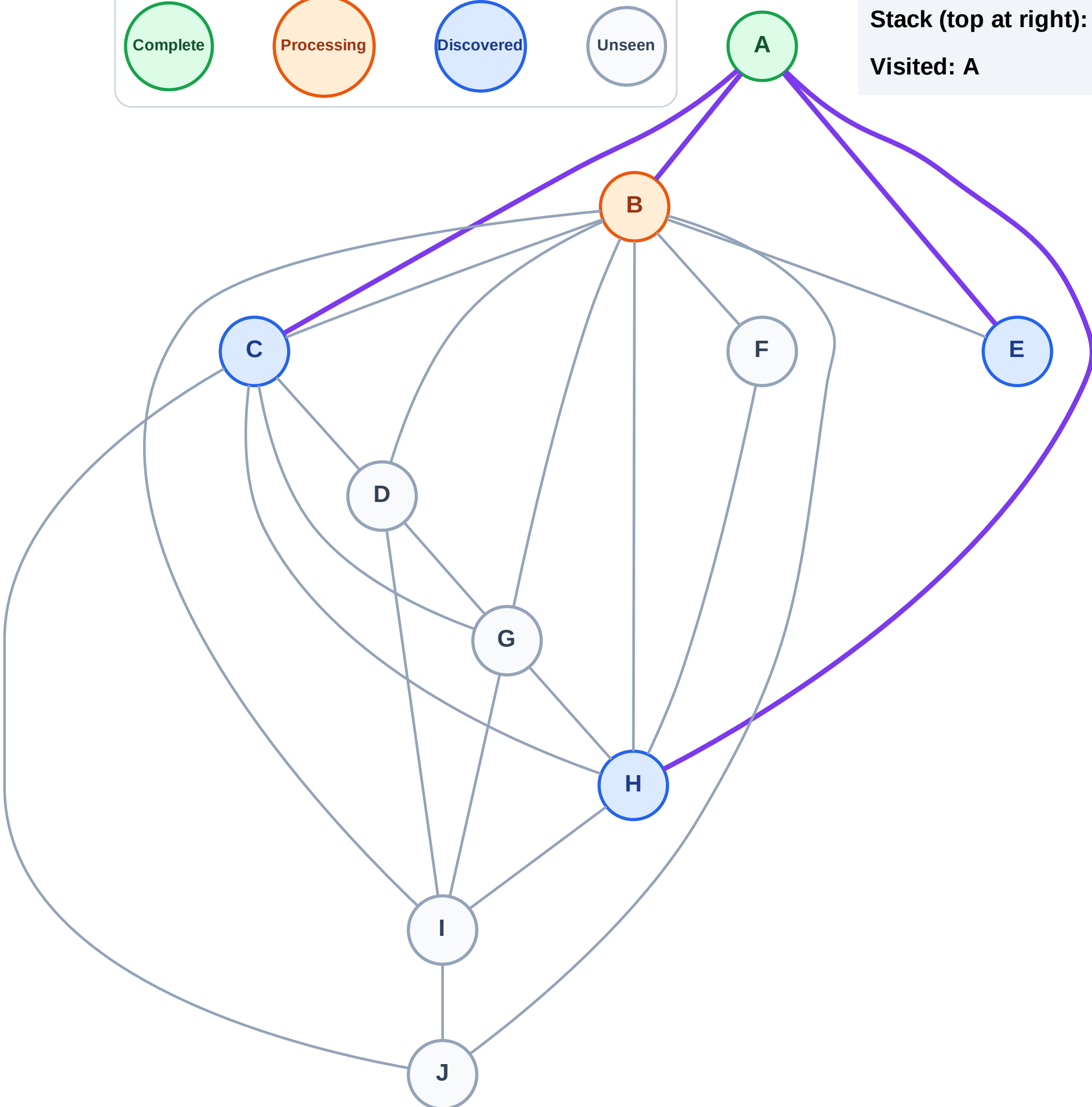
Step 4: Process node B

Legend



Stack (top at right): H | E | C

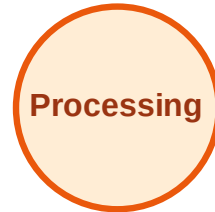
Visited: A



DFS Traversal

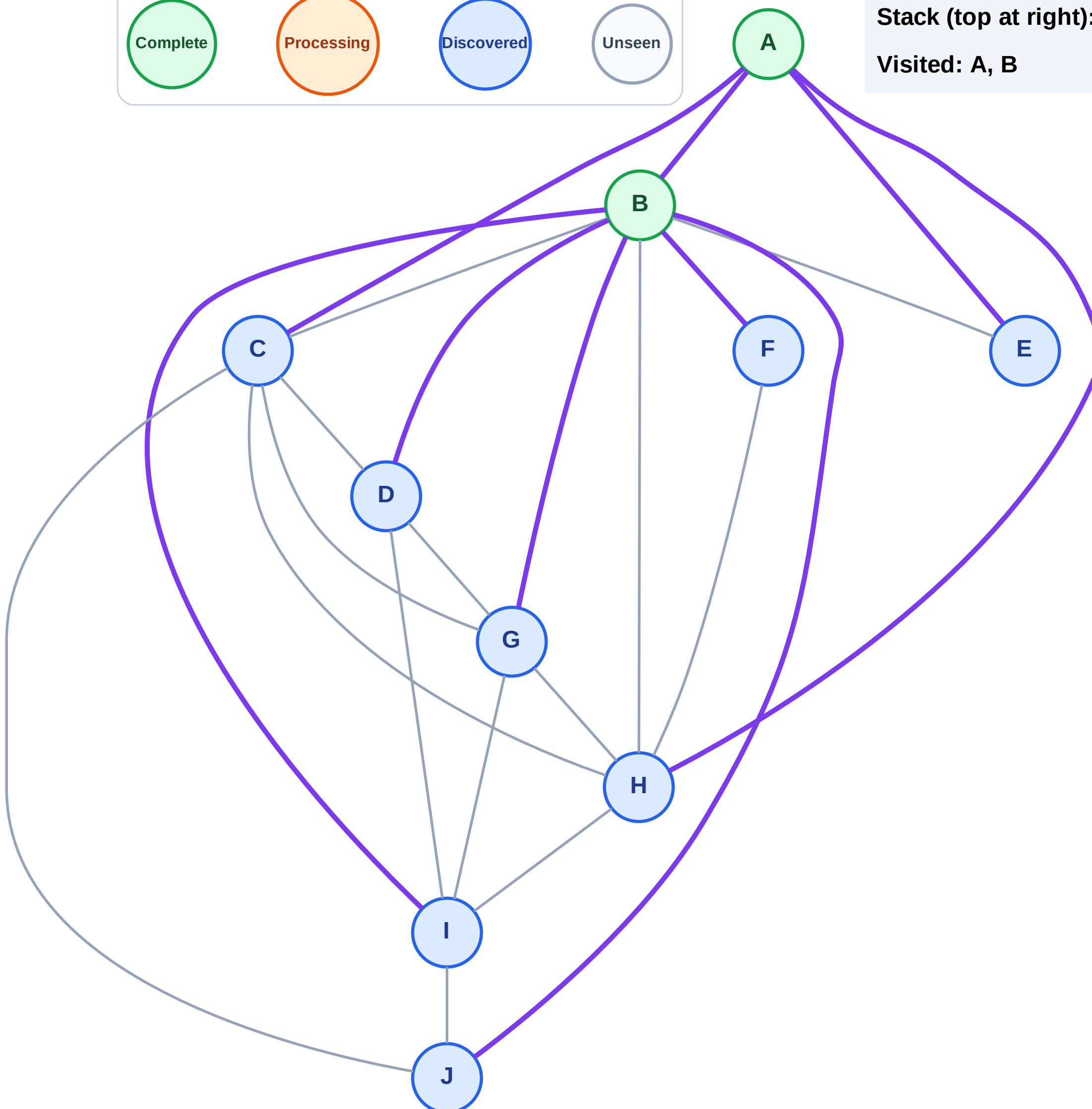
Step 5: Discover J, I, G, F, D from B

Legend



Stack (top at right): H | E | C | J | I | G | F | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

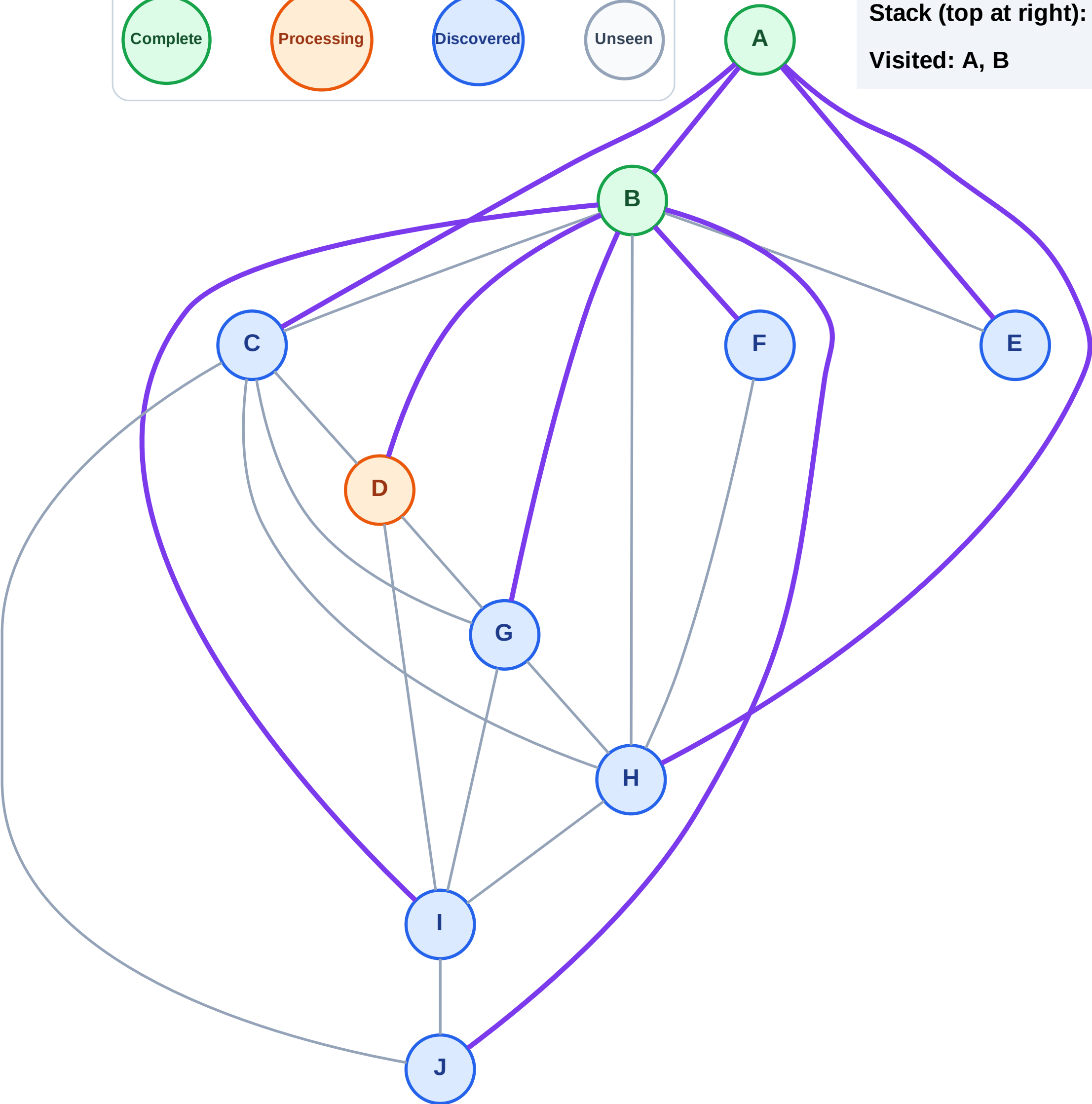
Processing

Discovered

Unseen

Stack (top at right): H | E | C | J | I | G | F

Visited: A, B



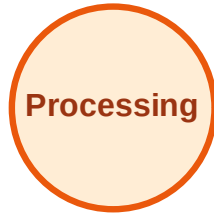
DFS Traversal

Step 7: No new neighbors from D

Legend



Complete



Processing



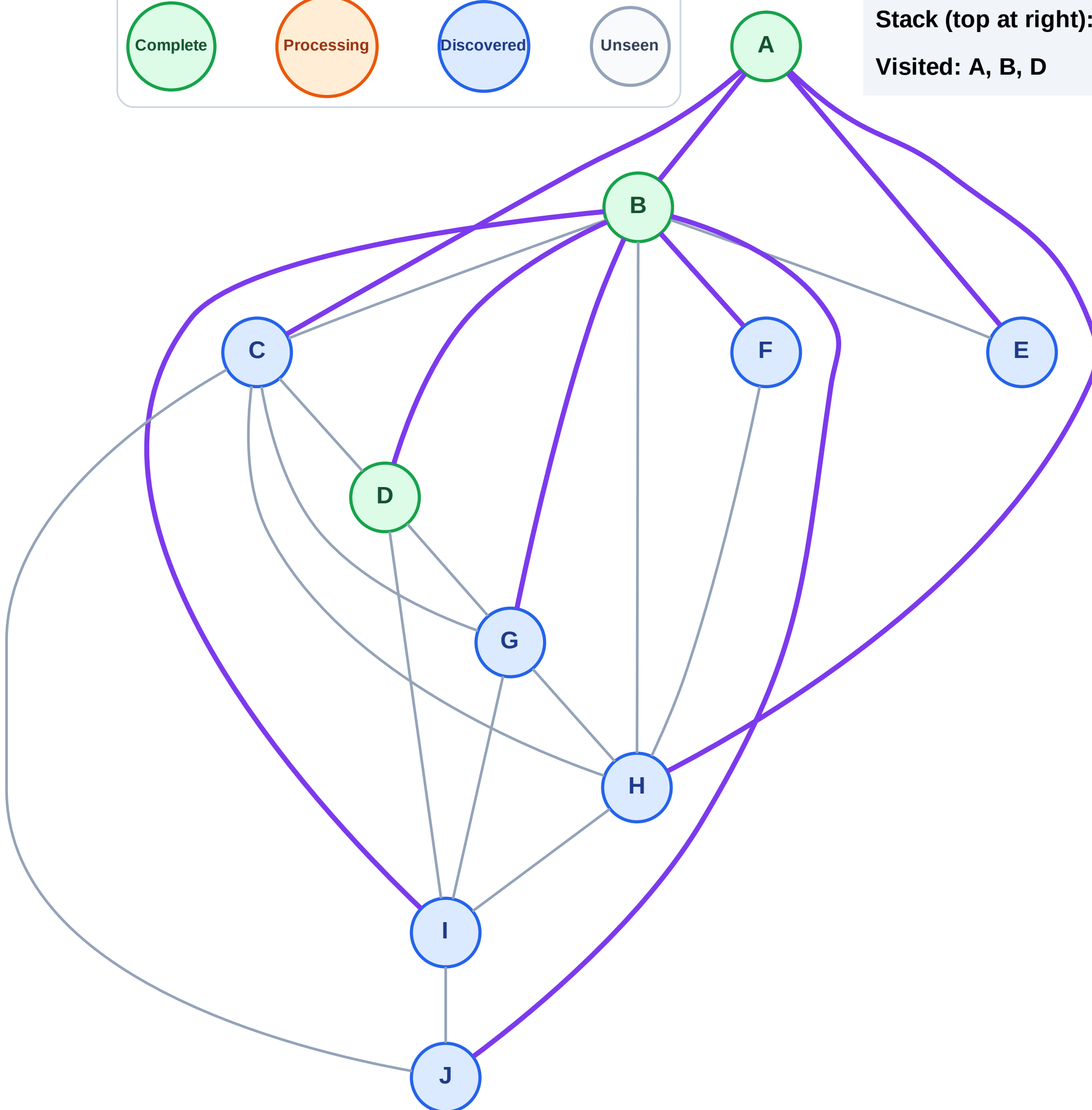
Discovered



Unseen

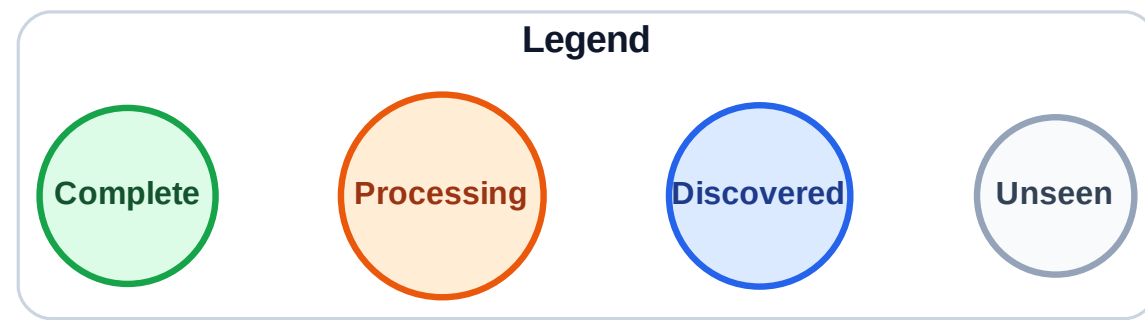
Stack (top at right): H | E | C | J | I | G | F

Visited: A, B, D



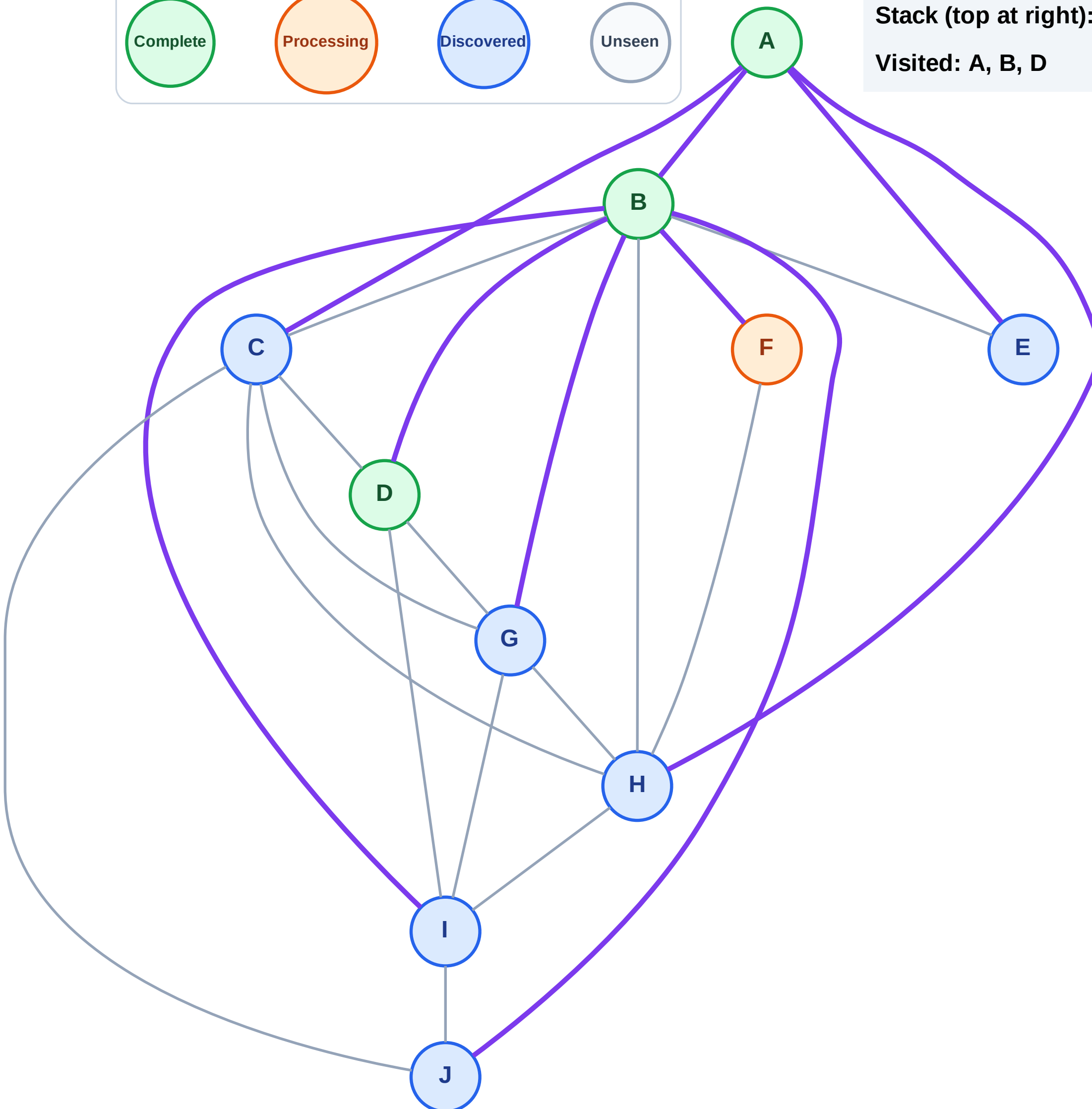
Step 8: Process node F

Page 10 of 10



Stack (top at right): H | E | C | J | I | G

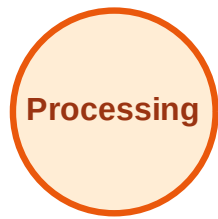
Visited: A, B, D



DFS Traversal

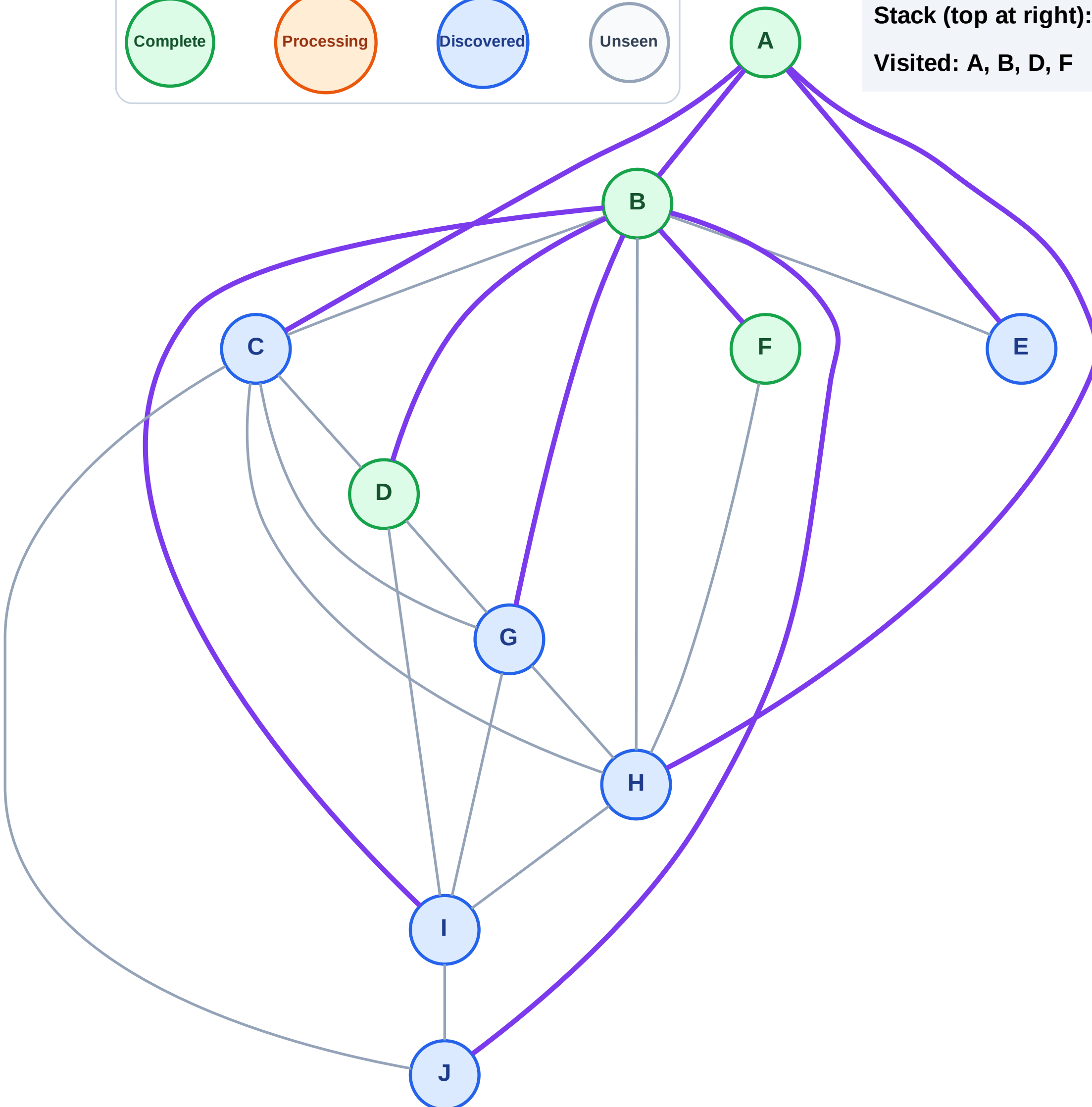
Step 9: No new neighbors from F

Legend



Stack (top at right): H | E | C | J | I | G

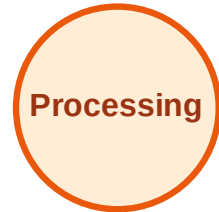
Visited: A, B, D, F



DFS Traversal

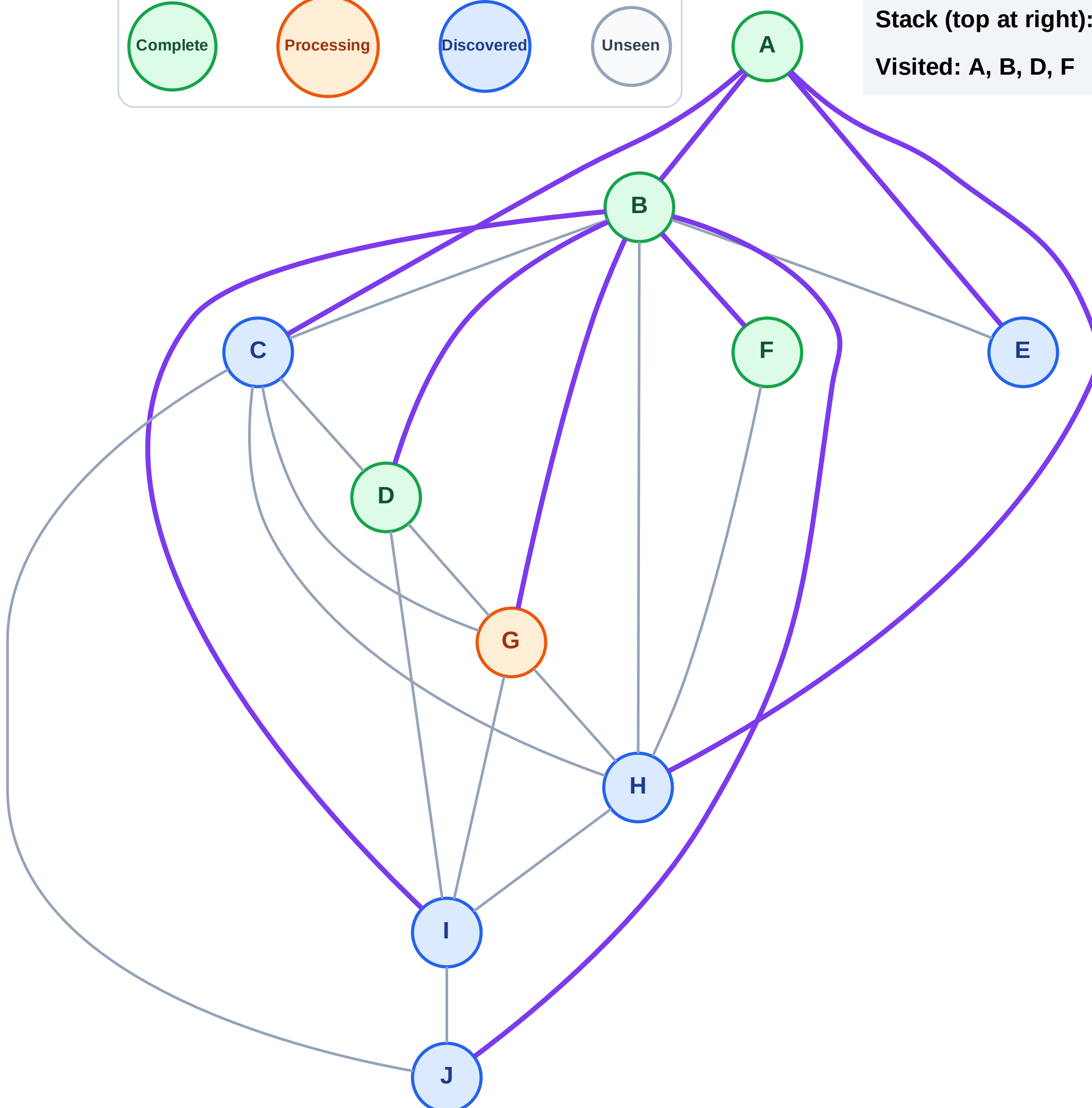
Step 10: Process node G

Legend



Stack (top at right): H | E | C | J | I

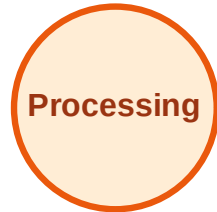
Visited: A, B, D, F



DFS Traversal

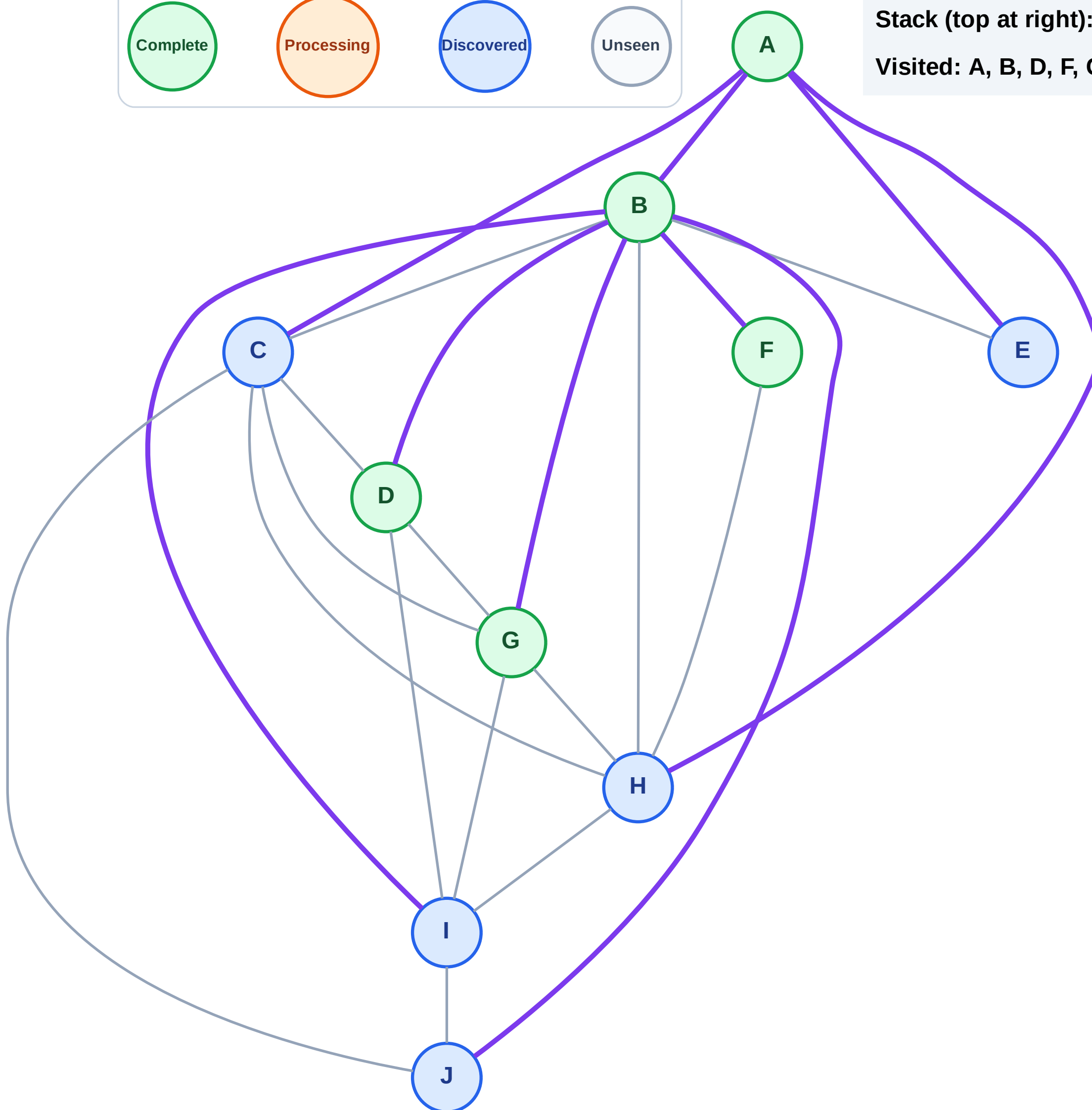
Step 11: No new neighbors from G

Legend



Stack (top at right): H | E | C | J | I

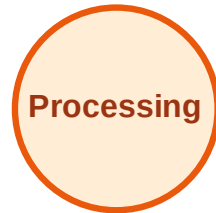
Visited: A, B, D, F, G



DFS Traversal

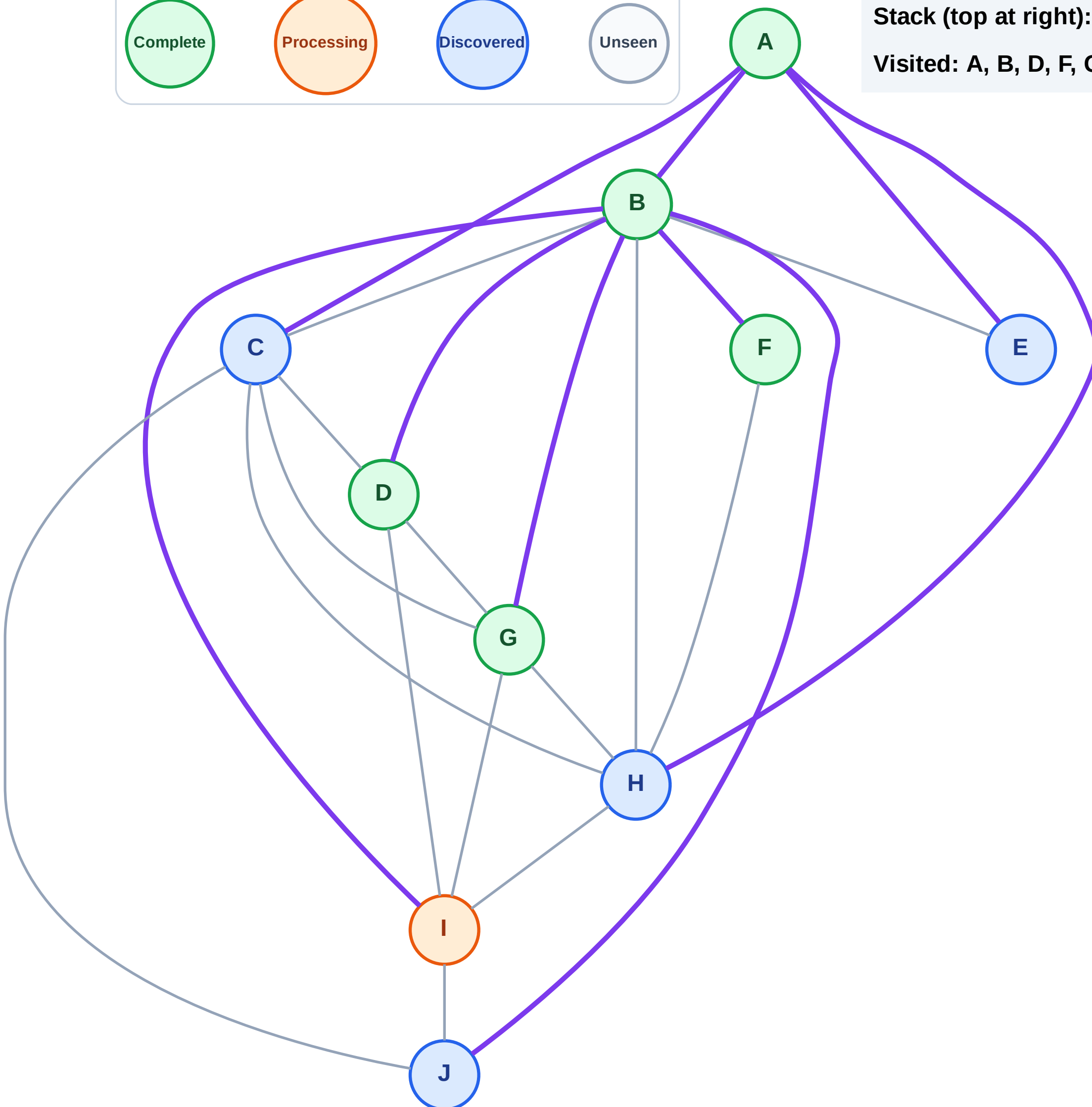
Step 12: Process node I

Legend



Stack (top at right): H | E | C | J

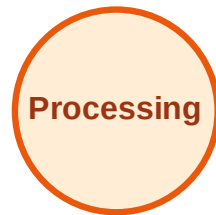
Visited: A, B, D, F, G



DFS Traversal

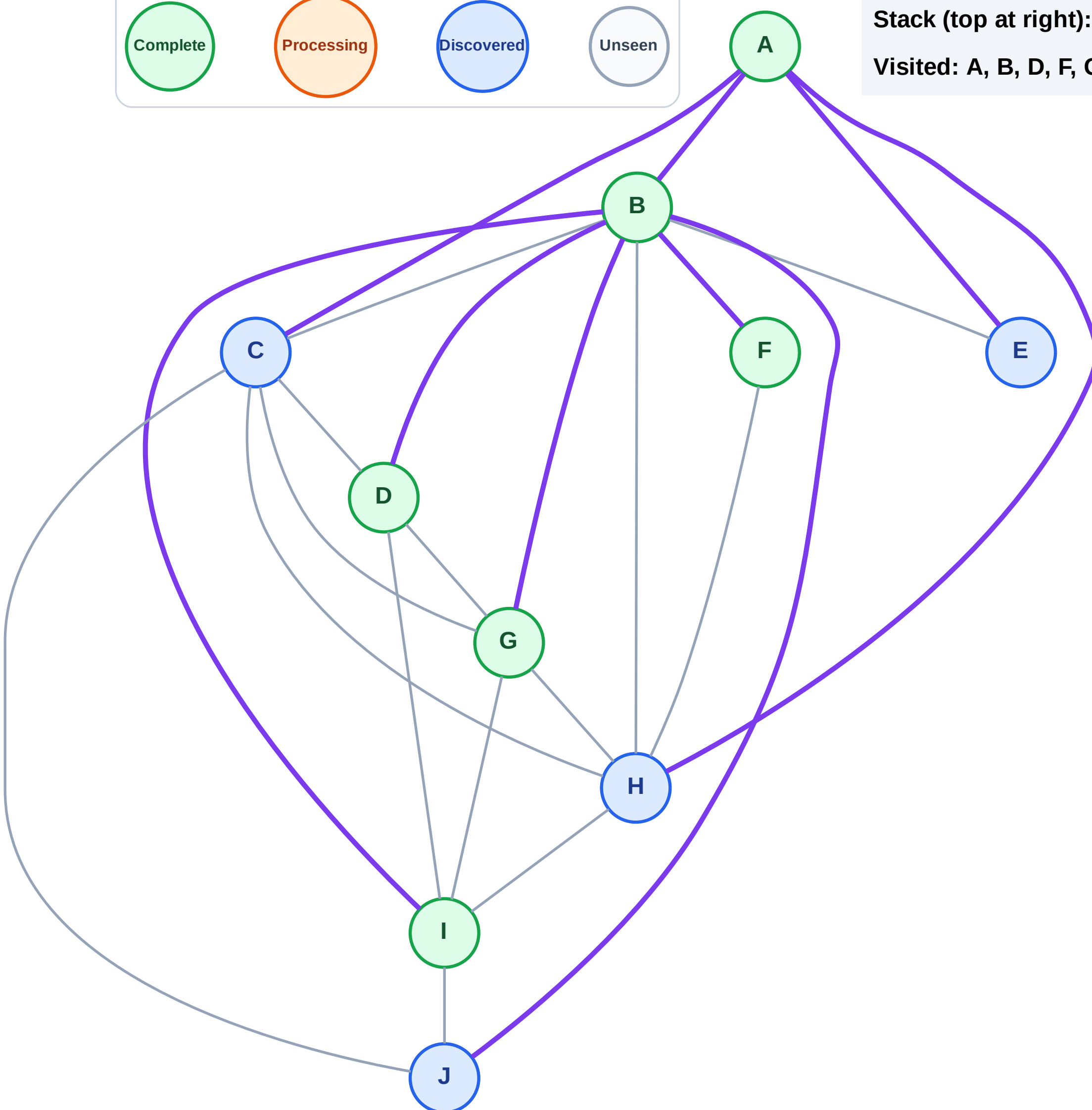
Step 13: No new neighbors from I

Legend



Stack (top at right): H | E | C | J

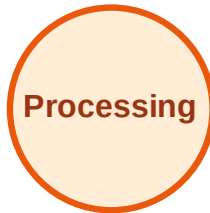
Visited: A, B, D, F, G, I



DFS Traversal

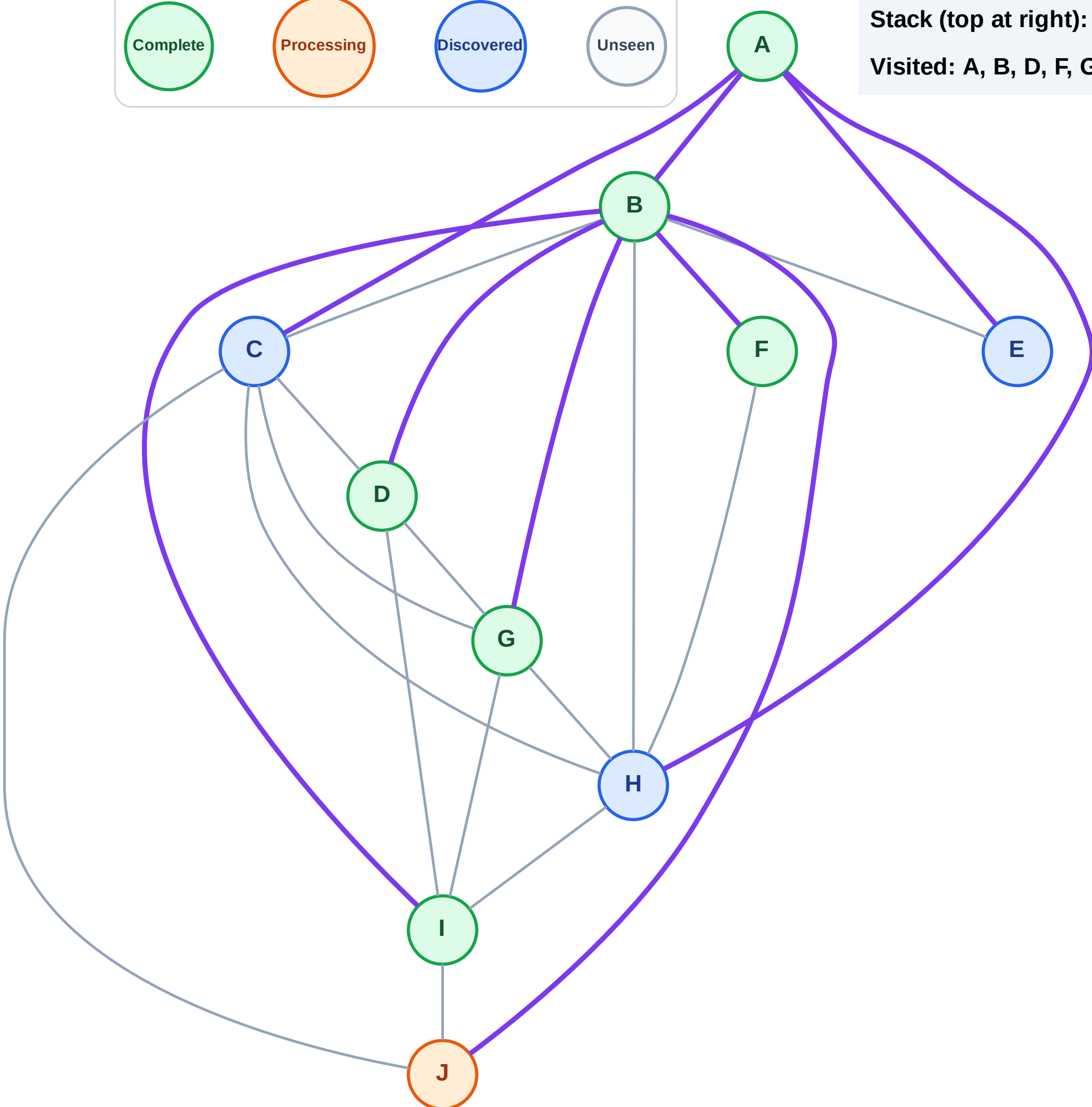
Step 14: Process node J

Legend



Stack (top at right): H | E | C

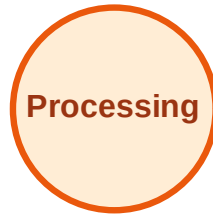
Visited: A, B, D, F, G, I



DFS Traversal

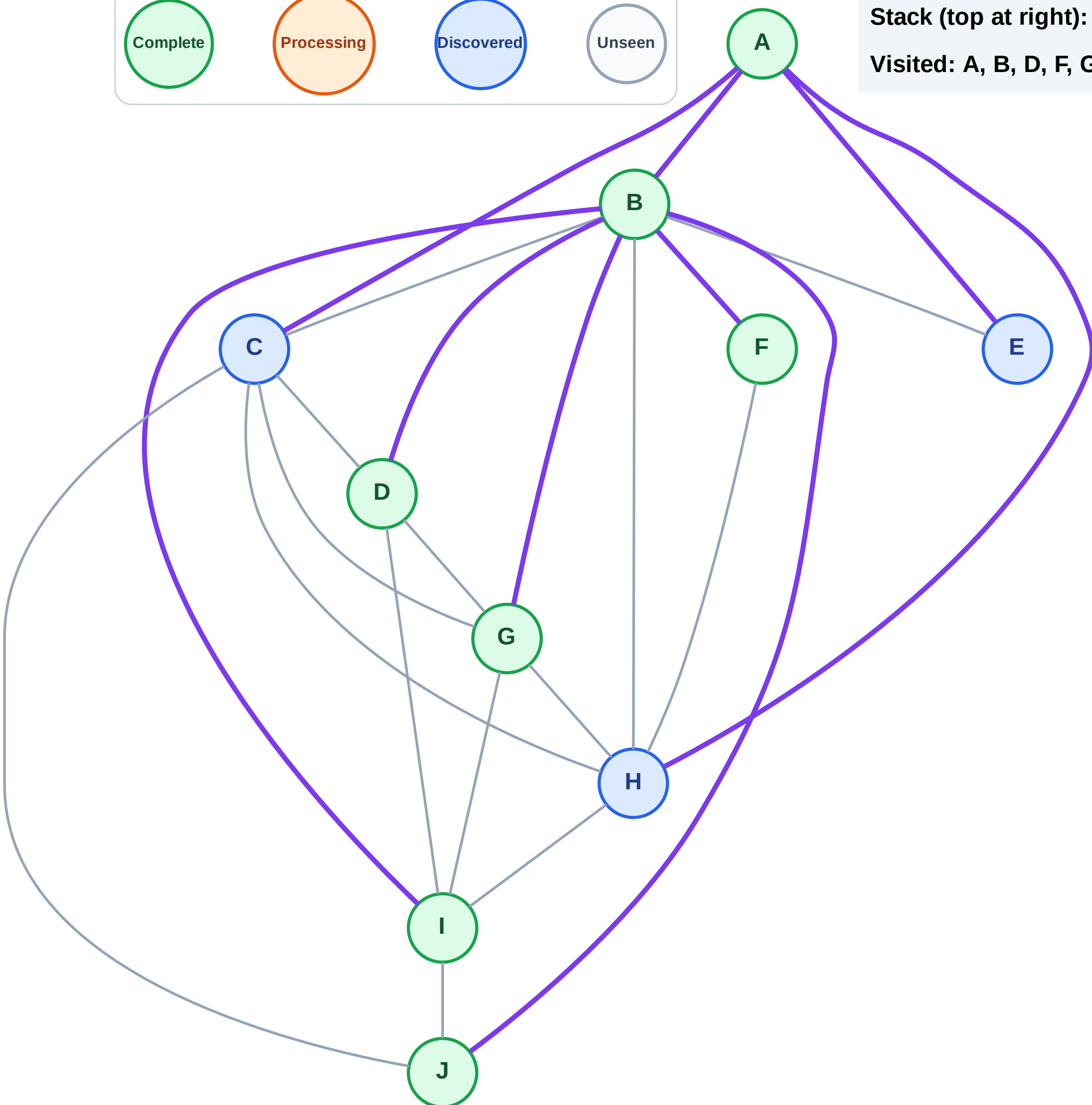
Step 15: No new neighbors from J

Legend



Stack (top at right): H | E | C

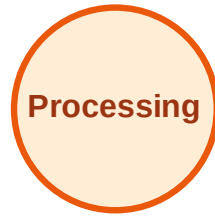
Visited: A, B, D, F, G, I, J



DFS Traversal

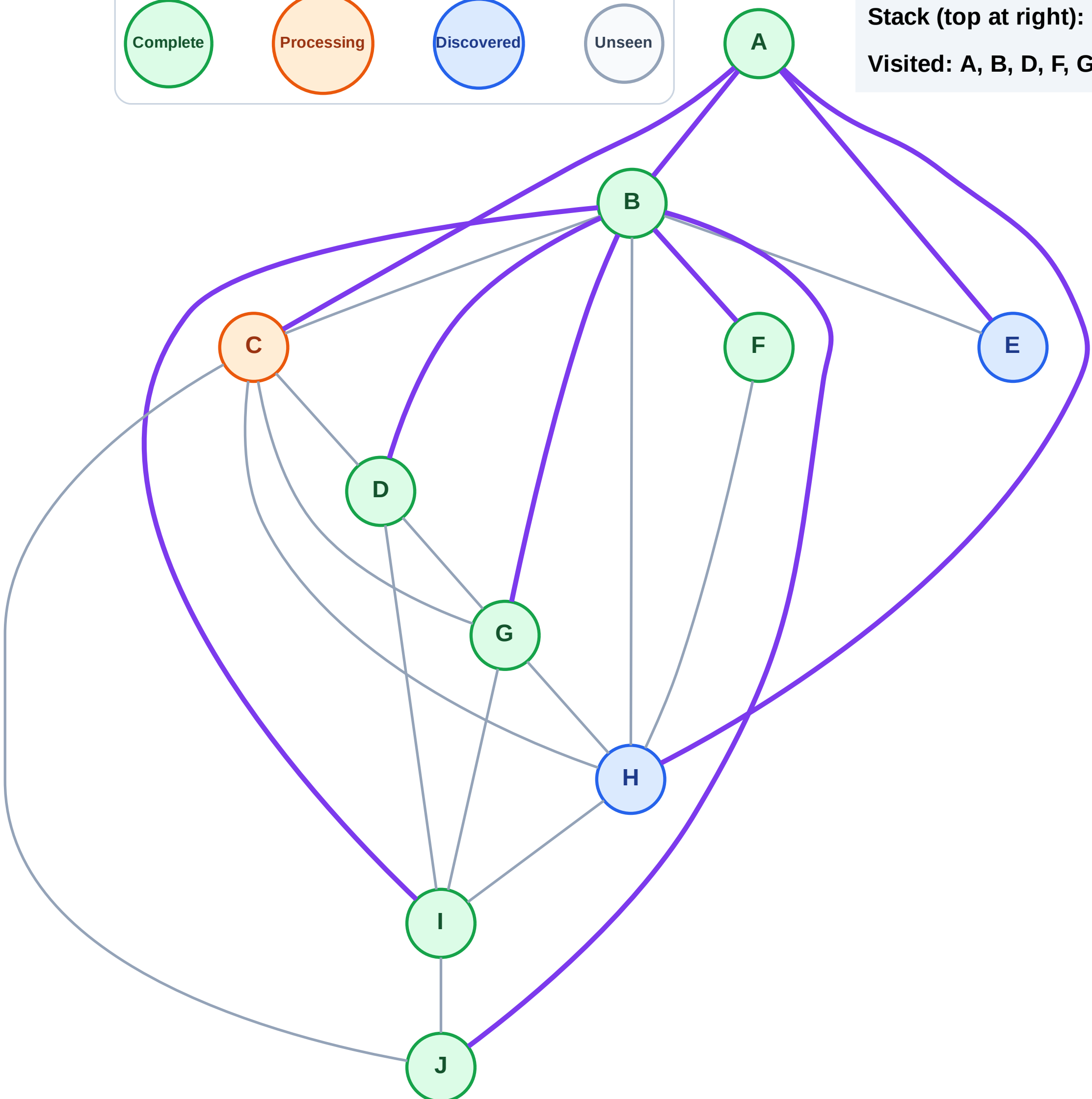
Step 16: Process node C

Legend



Stack (top at right): H | E

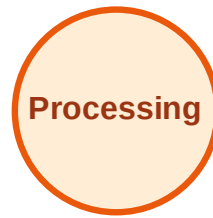
Visited: A, B, D, F, G, I, J



DFS Traversal

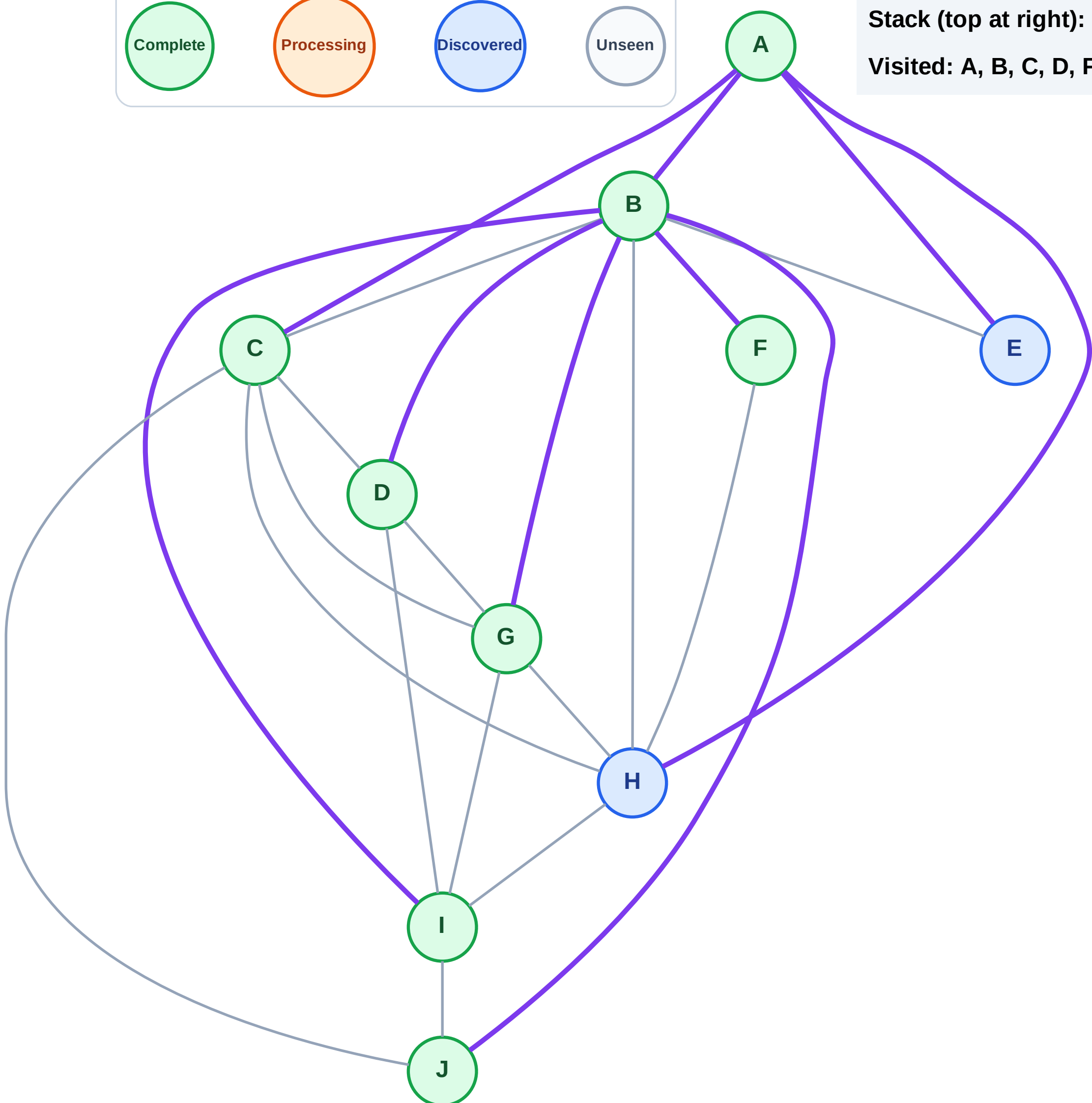
Step 17: No new neighbors from C

Legend



Stack (top at right): H | E

Visited: A, B, C, D, F, G, I, J



DFS Traversal

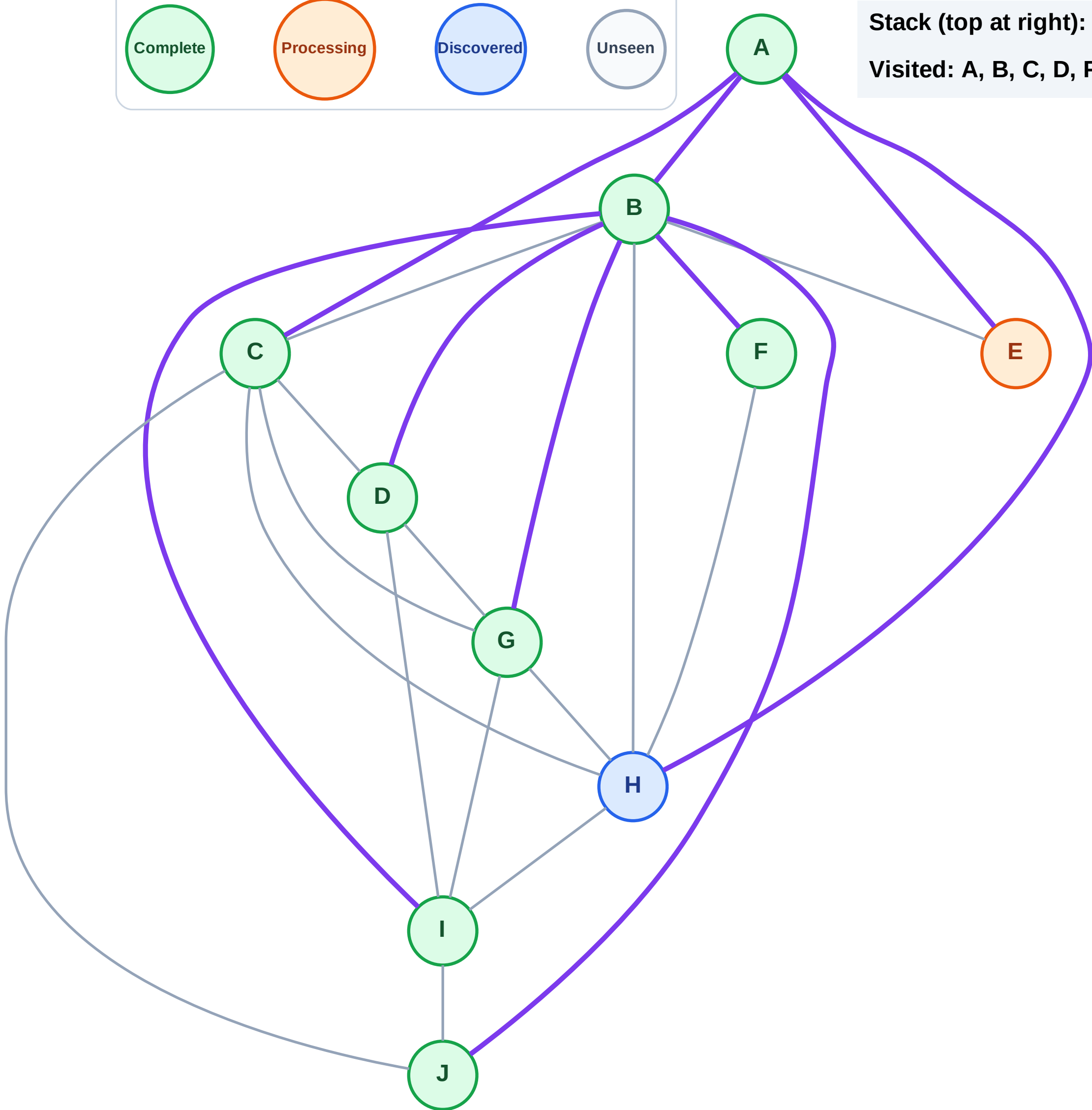
Step 18: Process node E

Legend



Stack (top at right): H

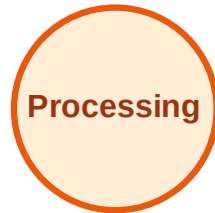
Visited: A, B, C, D, F, G, I, J



DFS Traversal

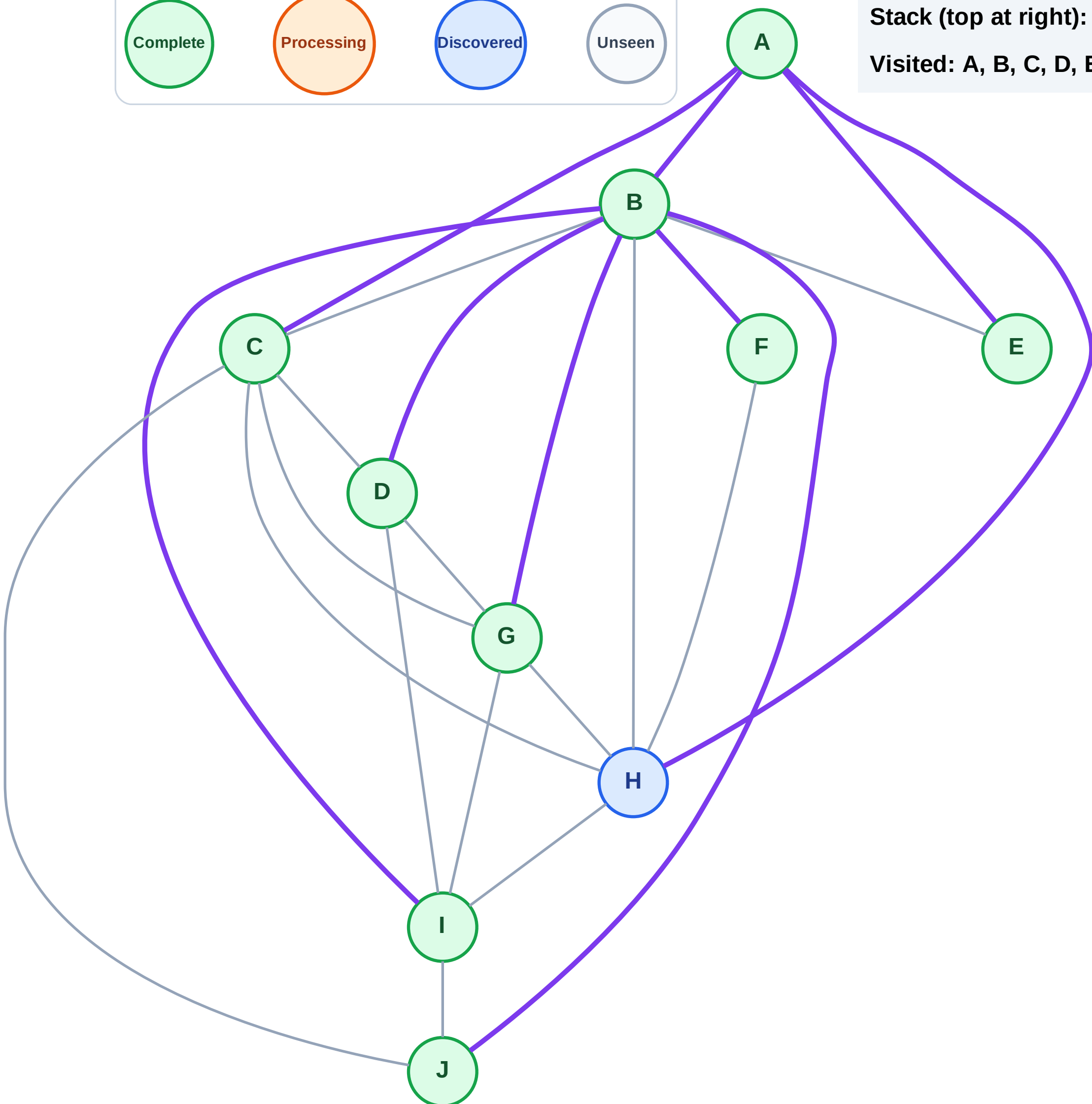
Step 19: No new neighbors from E

Legend



Stack (top at right): H

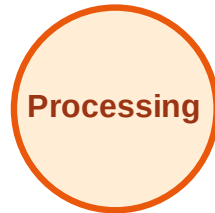
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

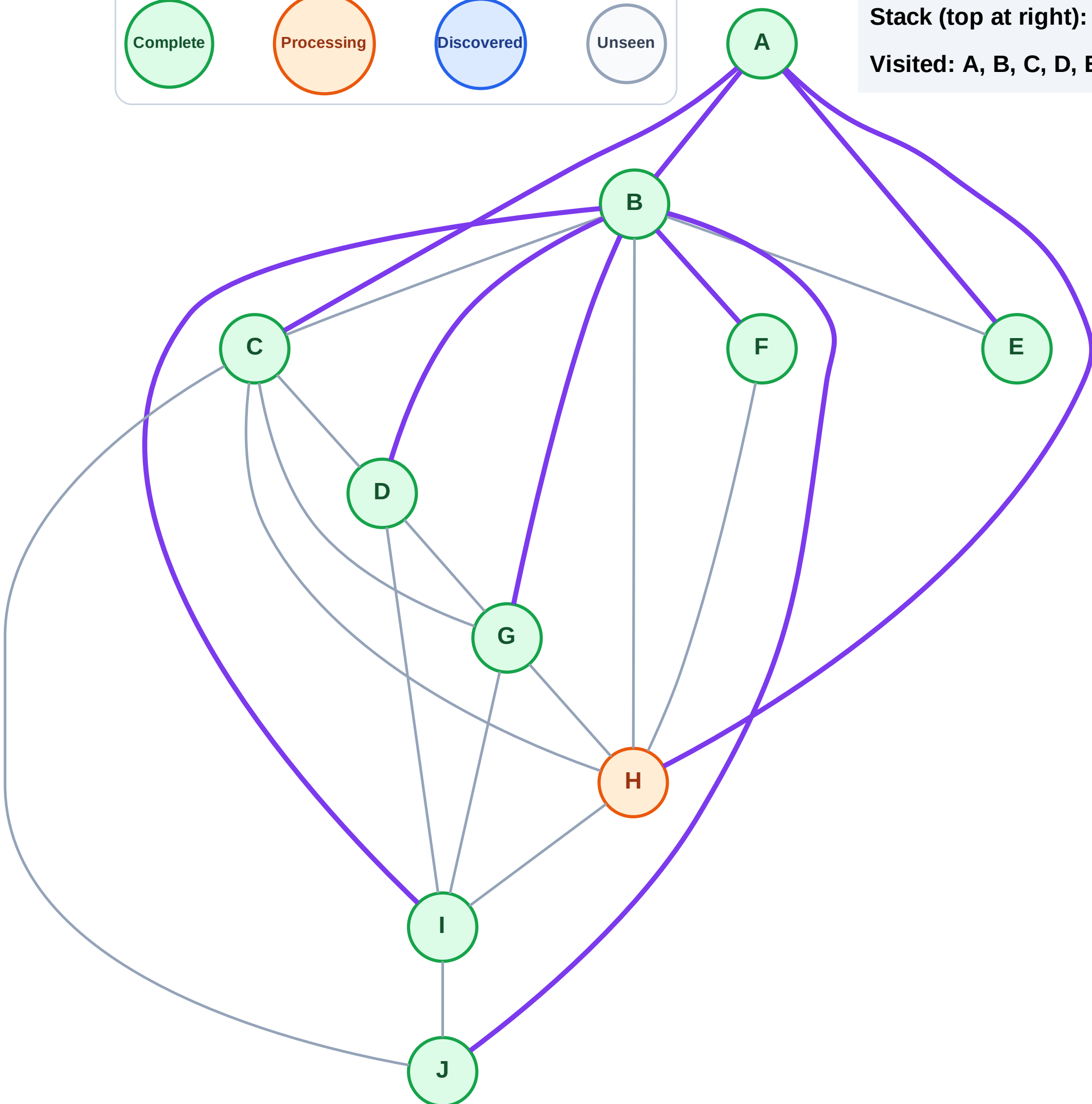
Step 20: Process node H

Legend



Stack (top at right): (empty)

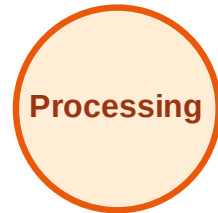
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

